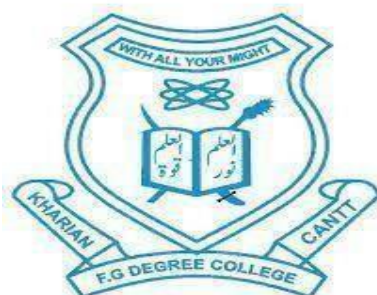




## **Final Year Design Project**

KIPS COACHING KHARIAN



**By**

**Raima Nasir                      2022-fkw-16**

**Maryam Iftikhar            2022-fkw-14**

**Kiran Shahzadi            2022-fkw-30**

**Nayab Zahoor              2022-fkw-15**

***Under the supervision of***

**Ms. Pakiza Saif**

**Session: 2022-2026**

**DEPARTMENT OF INFORMATION TECHNOLOGY  
F.G DEGREE COLLEGE FOR WOMEN, KHARIAN CANTT**

**FACULTY OF COMPUTING &  
INFORMATION TECHNOLOGY (FCIT), UNIVERSITY OF  
THE PUNJAB, LAHORE**



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**KIPS COACHING KHARIAN**

**A project presented to  
University of the Punjab, Lahore**

**In partial fulfilment of the requirement for the degree of**

**Session: 2022-2026**

**By**

|                        |                    |
|------------------------|--------------------|
| <b>Raima Nasir</b>     | <b>2022-FKW-16</b> |
| <b>Maryam Iftikhar</b> | <b>2022-FKW-14</b> |
| <b>Kiran Shahzadi</b>  | <b>2022-FKW-30</b> |
| <b>Nayab Zahoor</b>    | <b>2022-FKW-15</b> |

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INFORMATION TECHNOLOGY (FCIT),  
UNIVERSITY OF THE PUNJAB, LAHORE.**



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Signature: \_\_\_\_\_

Raima Nasir [080045]

Signature: \_\_\_\_\_

Kiran Shahzadi [080052]

Signature: \_\_\_\_\_

Maryam Iftikhar [080035]

Signature: \_\_\_\_\_

Nayab Zahoor [080039]



## CERTIFICATE OF APPROVAL

It is to certify that the final year design project (FYDP) of BSIT “Kips Coaching App Kharian” was developed by **MARYAM IFTIKHAR (2022-Fkw-14), NAYAB ZAHOOR (2022-FKW-15), RAIMA NASIR (2022-FKW-16) and KIRAN SHAHZADI (2022-FKW-30)**

Under the supervision of “Ms. PAKIZA SAIF” in my opinion it is fully adequate, in scope and quality for the degree of Bachelors of Science in INFORMATION TECHNOLOGY.

Signature: \_\_\_\_\_

Supervisor

Miss Pakiza Saif

FG Degree College for Women Kharian Cantt

Dated: \_\_\_\_\_

Signature: \_\_\_\_\_

HOD, Department of Information Technology

Mrs. Sidra Khan

FG Degree College for Women Kharian Cantt

Dated: \_\_\_\_\_



---

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It is certified that this document does not contain any spelling, punctuation or grammatical mistakes. This document is well organized and also meet the defined objectives.

| <b>Student Name</b> | <b>Registration<br/>Number</b> | <b>Roll no</b> | <b>Signature</b> |
|---------------------|--------------------------------|----------------|------------------|
| Raima Nasir         | (2022-fkw-16)                  | 080045         | _____            |
| Maryam Iftikhar     | (2022-fkw-14)                  | 080035         | _____            |
| Kiran Shahzadi      | (2022-fkw-30)                  | 080052         | _____            |
| Nayab Zahoor        | (2022-fkw-15)                  | 080039         | _____            |



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Their support helped us understand the requirements and build a better solution. Finally, we thank all those who helped us directly or indirectly and help us reach this stage.

Signature: -----

Raima Nasir [080045]

Signature: -----

Kiran Shahzadi [080052]

Signature: -----

Maryam Iftikhar [080035]

Signature: -----

Nayab Zahoor [080039]



## Abbreviations

|      |                                                 |
|------|-------------------------------------------------|
| FCIT | Faculty of Computing and Information Technology |
| JSON | JavaScript Object Notation                      |
| UI   | User Interface                                  |
| XML  | Extensible Markup Language                      |



## Table of Contents

|                                              |           |
|----------------------------------------------|-----------|
| <b>Introduction .....</b>                    | <b>13</b> |
| 1.Introduction .....                         | 14        |
| 1.1 Problem Statement .....                  | 14        |
| 1.2 Problem Solution .....                   | 14        |
| 1.3 Objectives of the Proposed System .....  | 14        |
| 1.4 Scope .....                              | 15        |
| 1.5 System Components .....                  | 15        |
| 1.6 Related Systems .....                    | 18        |
| 1.7 Vision Statement.....                    | 18        |
| 1.8 System Limitations and Constraints ..... | 19        |
| 1.9 Technologies and Tools .....             | 19        |
| 1.10 Project Deliverables .....              | 20        |
| 1.11 Project Planning .....                  | 21        |
| 1.12 Summary .....                           | 21        |
| <b>Literature Review .....</b>               | <b>22</b> |
| 2. Review.....                               | 23        |
| 2.1 Literature Survey.....                   | 23        |
| 2.2 Related Work .....                       | 23        |
| 2.3 Gap Analysis .....                       | 24        |
| 2.4 Summary .....                            | 24        |
| <b>Requirements Analysis.....</b>            | <b>25</b> |
| 3.Analysis.....                              | 26        |
| 3.1 User classes and Characteristics.....    | 26        |
| 3.2 Requirement Identifying Techniques ..... | 27        |
| 3.3 Functional Requirements.....             | 57        |
| 3.4 Non-Functional Requirements .....        | 71        |
| 3.5 External Interface Requirements .....    | 72        |
| 3.6 Summary.....                             | 72        |





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|                                          |           |
|------------------------------------------|-----------|
| <b>Design and Architecture.....</b>      | <b>75</b> |
| 4.System Design.....                     | 75        |
| 4.1 Design Consideration.....            | 75        |
| 4.2 Requirement Traceability Matrix..... | 75        |
| 4.3 Design Models.....                   | 77        |
| 4.4 Architectural Design .....           | 83        |
| 4.5 Data Design .....                    | 85        |
| 4.6 User Interface Design.....           | 88        |
| 4.7 Behavioral Model.....                | 91        |
| 4.8 Design Decisions.....                | 96        |
| 4.9 Summary.....                         | 98        |
| 4.10References.....                      | 98        |



## List of Figures

|                                             |    |
|---------------------------------------------|----|
| Figure 1: Gantt Chart .....                 | 21 |
| Figure 2: Use Case Diagram .....            | 28 |
| Figure 3: Story boarding .....              | 57 |
| Figure 4: Class Diagram .....               | 78 |
| Figure 5: Sequence Diagram (Admin).....     | 79 |
| Figure 6: Sequence Diagram (Student).....   | 80 |
| Figure 7: Sequence Diagram (Teacher) .....  | 81 |
| Figure 8: Sequence Diagram ( Parent).....   | 82 |
| Figure 9: State Transition Diagram.....     | 83 |
| Figure 10: Architecture Diagram .....       | 84 |
| Figure 11: Component Diagram .....          | 85 |
| Figure 12: ERD Schema .....                 | 86 |
| Figure 13:Screen Image .....                | 88 |
| Figure 14:Screen Image .....                | 89 |
| Figure 15:Screen Image .....                | 89 |
| Figure 16: Screen Image .....               | 90 |
| Figure 17:Sequence Diagram (Admin).....     | 92 |
| Figure 18: Sequence Diagram (Student).....  | 93 |
| Figure 19: Sequence Diagram (Teacher) ..... | 94 |
| Figure 20: Sequence Diagram ( Parent).....  | 95 |
| Figure 21: State Transition Diagram.....    | 96 |



## List of Tables

|                                                                            |    |
|----------------------------------------------------------------------------|----|
| Table 1: Related Systems.....                                              | 18 |
| Table 2: Technologies and Tools.....                                       | 19 |
| Table 3: Related Systems.....                                              | 23 |
| Table 4: User Classes and Characteristics .....                            | 26 |
| Table 5 Use Case 1: Register.....                                          | 29 |
| Table 6 Use Case 2: Reset password .....                                   | 30 |
| Table 7 Use Case 3: Login.....                                             | 31 |
| Table 8 Use Case 4:Update Profile.....                                     | 32 |
| Table 9 Use Case 5: Approve parent and student registration.....           | 33 |
| Table 10 Use Case 6: Add/Delete/update student teacher and parent .....    | 34 |
| Table 11 Use Case 7: Manage announcements.....                             | 35 |
| Table 12 Use Case 8: View system reports .....                             | 36 |
| Table 13 Use Case 9: Record fee payment .....                              | 37 |
| Table 14 Use Case 10: Classes timetable .....                              | 38 |
| Table 15 Use Case 11: View enrolled subjects.....                          | 39 |
| Table 16 Use Case 12: Attend online classes .....                          | 40 |
| Table 17 Use Case 13: Download study material.....                         | 41 |
| Table 18 Use Case 14:View and attempt online quizzess and assignment ..... | 42 |
| Table 19 Use Case 15: View fee history and status .....                    | 43 |
| Table 20 Use Case 16: Notification for fee payment.....                    | 44 |
| Table 21 Use Case 17: Create link for online classes.....                  | 45 |
| Table 22 Use Case 18:Upload study material.....                            | 46 |
| Table 23 Use Case 19:Upload and view assignment.....                       | 47 |
| Table 24 Use Case 20: Create online quizzess .....                         | 48 |
| Table 25 Use Case 21: Manage student perfomance .....                      | 49 |
| Table 26 Use Case 22: Communication with parents.....                      | 50 |
| Table 27 Use Case 23: Link parent to child.....                            | 51 |
| Table 28 Use Case 24: View child profile .....                             | 52 |
| Table 29 Use Case 25:View result .....                                     | 53 |
| Table 30 Use Case 26:Recive announcements .....                            | 54 |
| Table 31 Use Case 27: Feedback and review .....                            | 55 |
| Table 32 Use Case 28: Logout.....                                          | 56 |
| Table 33: Functional Requirement #1.....                                   | 59 |
| Table 34: Functional Requirement #2.....                                   | 59 |
| Table 35: Functional Requirement #3.....                                   | 60 |
| Table 36: Functional Requirement #4.....                                   | 60 |
| Table 37: Functional Requirement #5.....                                   | 60 |
| Table 38: Functional Requirement #6.....                                   | 61 |



|                                                                 |    |
|-----------------------------------------------------------------|----|
| Table 39: Functional Requirement #7.....                        | 61 |
| Table 40: Functional Requirement #8.....                        | 62 |
| Table 41: Functional Requirement #9.....                        | 62 |
| Table 42: Functional Requirement #10.....                       | 63 |
| Table 43: Functional Requirement #11.....                       | 63 |
| Table 44: Functional Requirement #12.....                       | 64 |
| Table 45: Functional Requirement #13.....                       | 64 |
| Table 46: Functional Requirement #14.....                       | 65 |
| Table 47: Functional Requirement #15.....                       | 65 |
| Table 48: Functional Requirement #16.....                       | 66 |
| Table 49: Functional Requirement #17.....                       | 66 |
| Table 50: Functional Requirement #18.....                       | 66 |
| Table 51: Functional Requirement #19.....                       | 67 |
| Table 52: Functional Requirement #20.....                       | 67 |
| Table 53: Functional Requirement #21.....                       | 68 |
| Table 54: Functional Requirement #22.....                       | 68 |
| Table 55: Functional Requirement #23.....                       | 69 |
| Table 56: Functional Requirement #24.....                       | 69 |
| Table 57: Functional Requirement #25.....                       | 70 |
| Table 58: Functional Requirement #26.....                       | 70 |
| Table 59: Functional Requirement #27.....                       | 71 |
| Table 60: Functional Requirement #28.....                       | 71 |
| Table 61: Traceability Matrix .....                             | 75 |
| Table 62: Data Dictionary .....                                 | 87 |
| Table 63: Screen object (Live classes).....                     | 90 |
| Table 64: Screen object (Notifications and announcements) ..... | 91 |
| Table 65: Screen object (Fee payment and confirmation) .....    | 91 |
| Table 66: Objects and Attributes .....                          | 97 |



## **Chapter 1**

### **Introduction**



## 1.Introduction

This chapter: Introduction contains a thorough overview, including goals, objectives and defined scope of our project. This chapter also represents the technologies and tools that will be used in implementation of project, along with Vision Statement. Importance of this chapter is that it fully explains the functionality and scope of proposed project. Achieving project objectives requires a complete understanding of the problem statement. Failure to do so will result in the objectives not being achieved.

### 1.1 Problem Statement

The existing system of **KIPS Coaching Kharian** is based on a manual process, which has led to several operational issues. Students face long queues during registration and fee submission due to the absence of an online payment facility. Additionally, there are no online scheduling updates, making the overall process time-consuming and inconvenient for students.

As a result, students are required to visit the coaching center repeatedly to obtain information, which leads to unnecessary time wastage. Teachers also experience difficulties in managing student records, sharing study materials and maintaining effective communication with parents. Furthermore, there is no centralized online platform that connects students, teachers and parents. In certain situations, such as adverse weather conditions like rain, students are unable to attend physical classes. This often results in missing important lectures and tests, further affecting their academic performance.

### 1.2 Problem Solution

The **KIPS Coaching Kharian** system is designed to address the challenges faced by students, teachers and administrators in managing academic activities. In the existing system, many tasks such as student registration and fee submission are performed manually, which leads to inefficiency and inconvenience.

This application provides a complete online platform where students can register, attend online classes, download study materials, check class schedules, pay fees and view results from anywhere. Teachers can conduct online classes, share notes and other learning resources and communicate effectively with students and parents. The administration can manage student records, monitor academic performance and handle the admission process online.

Overall, the **KIPS Coaching Kharian** system aims to make the learning process faster, more organized and more convenient for all users. It also supports students in different situations by providing uninterrupted access to educational resources.

### 1.3 Objectives of Proposed System

- To automate the Online Coaching service of KIPS.



- Live classes will conduct through the Google meet links.
- To provide features like online quizzes, result tracking and notification of class starting.
- To make communication between teachers and parents more effective.
- To ensure that students can access study material anytime and anywhere.
- To provide notifications to keep the students and teacher updated.
- To improve the design and layout for better user experience.
- To provide a better dashboard for monitoring student progress.
- To provide user friendly and accessible app for Android devices.

### 1.4 Scope

Our “**KIPS COACHING KHARIAN**” project is designed to modernize and enhance the learning mobile platform for students and teachers. Our project includes the following features:

#### **Project Scope**

- Develop a mobile-based digital learning platform for Matric and Intermediate students.
- Enable online enrollment and secure access for students, teachers and parents.
- Provide sharing of study materials in digital formats (PDFs, documents, images).
- Display student performance reports on a parent dashboard.
- Include messaging and notification features for effective communication.
- Focus only on digital learning features; no physical or classroom-based setup included.

### 1.5 System Components

System Components of KIPS COACHING KHARIAN are:

#### **Admin Module:**

##### **Login**

Admin securely log in to the system

Allow them to manage the whole platform

##### **Approve student and parent registration**

Review new register requests

Verify the details and approve or reject the request

Generate a unique ID for Student

##### **Add, update and delete student, teacher and parent**

Manually creates teacher accounts or edit information

Remove users from the system as needed

##### **Manage Announcements**

Post important updates

Announcements are visible to all users



### **View System Report**

Check reports

Help in monitoring and decision making

### **Record fee Payment**

Update the system when a student pays the fees

Ensure fee status is accurate and visible to the students and parents

### **Student Module:**

#### **Registration and Login Module**

Users can create an account (using email, phone number)

Existing users can log in securely

Password encryption and recovery options

#### **View Enroll subjects**

Check the subjects they enrolled

Students can enroll more than one subject.

#### **Attend online Classes**

Students take online classes as per schedule

Classes schedule or time table will be shared with students

#### **Download study material**

Students access the study material uploaded by the teacher

Material can be saved for the offline study

#### **View and attempt online quizzes and assignments**

Students can view assignments.

Students take quizzes created by teachers.

Scores may be automatically calculated after submission.

#### **View results, fee payment and fee history status**

Students can view their academic results.

They also check fee status and payment history anytime.

#### **Receive Notifications for fee payment**

The student receives instant notifications for fee payment status and due alerts, including paid amount and balance

#### **Receive Announcements**

Students receive alerts for new quizzes, assignments, announcements or fees.

Helps them stay updated

### **Teacher Module:**





### **Login**

Teachers securely log in to manage academic tasks.

Login provides access to teacher-specific tools.

### **Take Classes and Upload study material**

Teachers will take classes as per schedule and upload notes, slides and resources for students.

Files become instantly available to enrolled subjects.

### **Create quizzes**

Teachers create quizzes and set questions.

Students can attempt them online through the system.

### **Create and share link for online classes**

Teacher creates a link for online classes.

After creating the link, teacher share this link with the students.

### **Manage student performance**

Teachers view student progress.

They update marks or give feedback accordingly.

### **View and upload assignments**

Teacher upload assignment for students to complete.

They also review submitted student work and update status.

### **Announcements**

View Notifications

### **Response and communication with parents**

Teachers can message to parents regarding student performance or concern.

Help to build strong communication.

## **Parent Module:**

### **Registration and login**

Parents create account and log in.

Login ensure secure access to child-related information.

### **Link parent to child account**

Parents connect their account with their child's profile.

This give them visibility to academic and fee details.

### **View results**

Parents can see the child's marks, performance reports and progress.

Help them to track academic improvement.

### **Fee payment**

Parents can pay the child's fee through the system.



Payment status update immediately.

#### **Fee history status**

Parents check past payment and pending dues.

Provide complete transparency of financial records

## 1.6 Related systems

There are similar applications related to educational management systems that are functioning effectively in the education industry. These include KIPS Virtual App, The Educator Online System and Roots International Learning Platforms. These institutions have successfully implemented digital platforms and achieved notable success in managing academic and administrative operations. But these systems do not provide fully functional system for live classes and not even any dashboard where student and parents can check/view progress report. KIPS Virtual App just provide video lecture. Due to these limitations, our project aims to create a complete online education system where students can attend live classes, access study material and attend classes from home. Additionally, parents will be able to communicate with teachers and monitor their child's academic performance.

*Table 1: Related Systems*

| <b>App name</b>                       | <b>Weakness</b>                                                                                                                                                                                                                                          | <b>Proposed Solutions</b>                                                                                                                                               |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| KIPS Virtual App                      | The KIPS Virtual System provides limited access for students, as it primarily focuses on student enrollment and recorded video lectures. There is no facility for live online classes which reduces real-time interaction between students and teachers. | An integrated system is proposed that provides online classes, student performance tracking and announcement features to improve communication and academic management. |
| Roots International Learning Platform | High subscription and maintenance cost.                                                                                                                                                                                                                  | An affordable and scalable solution is proposed for medium-sized academies like KIPS Coaching Academy, reducing costs while supporting future growth.                   |

## 1.7 Vision Statement



Students at **KIPS Coaching Kharian** currently face several challenges due to the manual system, including long queues for fee submission, lack of timely updates and the need to physically visit the campus to obtain information. To address these issues, the (App) **KIPS Coaching Kharian** is being developed to provide a digital solution that offers simple registration, online fee payment, instant updates, notifications and easy access to class schedules and other important information from anywhere. This application aims to create a more efficient, accessible and student-friendly learning environment.

## 1.8 System Limitations and Constraints

### Limitations

The main limitation of our project is that **KIPS Coaching Kharian** serves only for the Kharian and surrounding. Our app is specifically designed for students of this branch. Another limitation is that the app currently covers only Punjab region data and does not include information for other KIPS branches across Pakistan. The app needs an active internet connection to work properly.

### Constraints

The app does not support offline mode, so user must be connected to the internet to access features. The **KIPS COACHING KHARIAN** is only android app.

## 1.9 Technologies and Tools

*Table 2: Technologies and Tools*

| Tools, Libraries and Technology | Tools          | Version | Rationale                                              |
|---------------------------------|----------------|---------|--------------------------------------------------------|
|                                 | Android studio | Narwhal | Used for design, code, test and debug                  |
|                                 | Libraries      | Version | Rationale                                              |
|                                 | Firebase       | 3.x     | Cloud-based database management system                 |
|                                 | Copyleaks      | Latest  | Used for plagiarism detection and AI generated content |



|  |                   |                |                                                                      |
|--|-------------------|----------------|----------------------------------------------------------------------|
|  |                   |                | analysis                                                             |
|  | <b>Technology</b> | <b>Version</b> | <b>Rationale</b>                                                     |
|  | Java              | 8              | Programming language for building: backend and logic of application  |
|  | JSON              | N/A            | Used for structuring all data sent to and received from Firebase     |
|  | XML               | 1.0            | For designing the interface of the app                               |
|  | Node.js           | N/A            | For server-side logic, Firebase functions and payments gateway setup |

## 1.10 Project Deliverables

### Project Deliverables:

**Project Proposal:** initial document covering basic detail of the project i.e., scope, objectives and planning.

**SRS (Software Requirement Specification):** This includes all the functional and non-functional requirements that the project will cover.

**Design Specification Document:** Includes the overall system design and architecture, including User Interfaces and data design as well.

**Prototypes:** Dummy working project.

**Final Project:** Fully developed mobile application of our Service.

**Admin Panel:** A panel/dashboard for the admin to manage and view details.

**Testing Reports:** Outlining test cases and their results.



**Project Report:** Complete FYP documentation enlisting all the specifications of our system.

**Presentation:** A final presentation of our application.

### 1.11 Project Planning



Figure 1: Gantt Chart

### 1.12 Summary

This chapter introduces the functionalities and core features of our mobile application. First, it states the problem that led to the development of our mobile application followed by the solution which is our app named **KIPS Coaching Kharian**. This chapter also outlines the goals and objectives of our proposed system, along with the risks, limitations and dependencies in details. The scope of our project is included and the system components are clearly defined. A literature review is provided to analyze related systems. We crafted a vision statement that conveys the long-term purpose, intent and significance of our product. Additionally, this chapter list the limitations and constraints of our system and describe the technologies and tools used in the project's implementation. To represent plans for carrying out project activities and showcasing milestones, a Gantt chart is also displayed.



## **Chapter 2**

### **Literature Review**



## 2.Review

Literature Review of our project highlights the previous researches, systems and technologies related to our system. This review helps by providing the modern recommendations and upgrade the system.

### 2.1 Literature Survey

Recent studies on Learning management system (LMS) shows that digital platforms improve learning efficiency by offering online content, quizzes, scheduling and progress tracking. Research highlights that mobile based system are more effective in developing countries because they reduce physical barriers such as distance and weather etc. For such system to succeed features like user authentication, content management and fee deposits are essential. However, challenges such as poor internet access, limited teacher training and lack of integrated systems for students, parents and teachers still affect adoption in many educational institutions.

### 2.2 Related work

As there is a manual system, there is no online system of **KIPS Coaching Kharian**. However, there are similar applications related to educational management systems that are functioning effectively in the education industry. These include KIPS Virtual App, The Educators Online System and Roots International Learning Platforms. These institutions have successfully implemented digital platforms and achieved notable success in managing academic and administrative operations.

*Table 3: Related Systems*

| App name         | Weakness                                                                                                                                                                                                                                                 | Proposed Solutions                                                                                                                                                      |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| KIPS Virtual App | The KIPS Virtual System provides limited access for students, as it primarily focuses on student enrollment and recorded video lectures. There is no facility for live online classes which reduces real-time interaction between students and teachers. | An integrated system is proposed that provides online classes, student performance tracking and announcement features to improve communication and academic management. |



|                                       |                                         |                                                                                                                                                       |
|---------------------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Roots International Learning Platform | High subscription and maintenance cost. | An affordable and scalable solution is proposed for medium-sized academies like KIPS Coaching Academy, reducing costs while supporting future growth. |
|---------------------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|

### 2.3 Gap analysis

The comparison of existing system shows several gaps that **KIPS COACHING KHARIAN** will address:

Lack of integrated Student, Teacher and Parent communication in current systems.

No user friendly branch specific app managing registrations, schedules, study material and fee payments.

Existing systems does not solve problems like online classes, registration or communication with parents about student performance.

Many platforms lack low cost online payment integration, which your system introduce.

### 2.4 Summary

The literature and related work indicate a strong need for digital academic management systems that are affordable, mobile friendly and suitable for institution. Current portal provides partial digital services but fail to address issues like communication gaps, accessibility and manual work load. Our system provides a digital platform for students, teacher, parent and administrator that supports efficient learning and management.





## **Chapter 3**

### **Requirements Analysis**



### 3. Analysis

Software requirement specification for our project SRS, which addresses both the functional And non-functional needs of our mobile application, is presented in this chapter. Along with the hardware, software and communication interfaces that will be utilized to carry out this project. It also outlines the requirements for the user and external interfaces. This chapter acts as a point of reference for the project's progress.

#### 3.1 User Classes and Characteristics

*Table 4: User Classes and Characteristics*

| User Class | User Characteristics                                                                                                                                                                                                                                                                                |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Admin      | Secure login<br>View and manage all users<br>Add, Update and remove user's account<br>Generate unique ID for Student<br>Create classes timetable<br>Assign teacher to specific classes<br>View all scheduled sessions and Enrollments<br>Enrolled Student report<br>Send announcements/Notification |
| Student    | Users can create an account<br>Edit personal details<br>View enrolled subjects and upcoming sessions<br>Attend online classes<br>Browse subjects offering by institute<br>View and attempt quizzes and assignments<br>Receive alerts for upcoming sessions and assignments/quizzes deadlines        |
| Teacher    | Login and profile verification<br>Join classes and deliver lecture<br>Upload study material<br>Create and share link for classes<br>View enrolled students per subjects                                                                                                                             |



|        |                                                                                                                                                                   |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        | Create quizzes and assignments<br>Upload assignment/quizzes marks                                                                                                 |
| Parent | Registration and login<br>Communication with teachers directly<br>Access Child's dashboard<br>Online fee payment<br>Receive instant alerts<br>Review and Feedback |

### 3.2 Requirement Identifying Techniques

This part enlists various techniques that are used to identify the requirements of our project **KIPS COACHING KHARIAN**.

#### Use Case Diagram

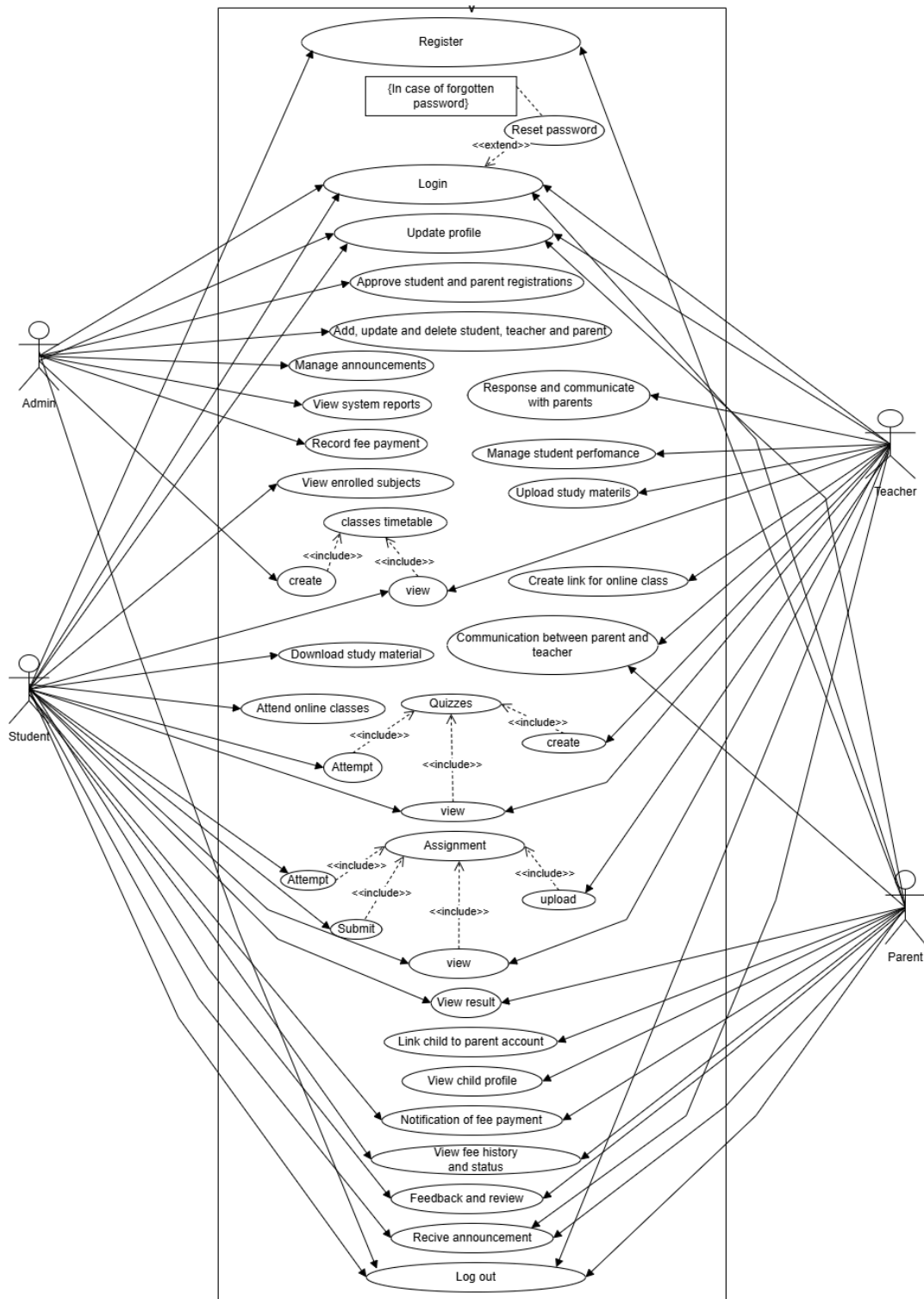


Figure 2: Use case diagram

*Table 5 Use Case 1: Register*

|                              |                                                                                                                                                                                                                                 |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b>                   | UC#1                                                                                                                                                                                                                            |
| <b>Title</b>                 | Register                                                                                                                                                                                                                        |
| <b>Description</b>           | Allow users (student and parent) to register with the KIPS Coaching Kharian app                                                                                                                                                 |
| <b>Primary Actors</b>        | Student, Parent                                                                                                                                                                                                                 |
| <b>Preconditions</b>         | Students and Parents are not already registered and has valid username /email.                                                                                                                                                  |
| <b>Post conditions</b>       | User account is created and stored in the system database.                                                                                                                                                                      |
| <b>Main success scenario</b> | Users (student and parent) selects “Register”.<br>Users (student and parent) enters required details (Full name, Email, username, Password).<br>After validation, system successfully registers the users (student and parent). |
| <b>Alternate Flow</b>        | Already registered Login                                                                                                                                                                                                        |
| <b>Exception</b>             | Invalid email format and weak password                                                                                                                                                                                          |
| <b>Status</b>                | Completed                                                                                                                                                                                                                       |
| <b>Priority</b>              | Very High                                                                                                                                                                                                                       |
| <b>Include</b>               | Validate user input                                                                                                                                                                                                             |



*Table 6 Use Case 2: Reset Password*

|                              |                                                                                                                                                                                                                                                                                         |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b>                   | UC#2                                                                                                                                                                                                                                                                                    |
| <b>Title</b>                 | Reset Password                                                                                                                                                                                                                                                                          |
| <b>Description</b>           | Allows a registered user to reset their forgotten password for the KIPS COACHING KHARIAN APP.                                                                                                                                                                                           |
| <b>Primary Actors</b>        | Student, Admin, Parent                                                                                                                                                                                                                                                                  |
| <b>Preconditions</b>         | The users (student, admin and parent) must have a registered account within the system                                                                                                                                                                                                  |
| <b>Post conditions</b>       | The user's password is successfully reset and they can access their account with the new credentials.                                                                                                                                                                                   |
| <b>Main success scenario</b> | Users select forgotten password or reset password. User enters their registered email or mobile number. After validation the system sends a password reset link or temporary password. User follow the instructions to set a new password. System successfully updates a user password. |
| <b>Alternate flow</b>        | User enters an unregistered email.<br>Password reset link expires<br>User requests multiple password resets.                                                                                                                                                                            |
| <b>Exception</b>             | System failure during password update.<br>Emails delivery failure<br>Invalid password format.                                                                                                                                                                                           |
| <b>Status</b>                | Proposed                                                                                                                                                                                                                                                                                |
| <b>Priority</b>              | High                                                                                                                                                                                                                                                                                    |
| <b>Include</b>               | Validate user input                                                                                                                                                                                                                                                                     |



*Table 7 Use Case 3: Login*

|                              |                                                                                                                                                                                |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b>                   | UC#3                                                                                                                                                                           |
| <b>Title</b>                 | Login                                                                                                                                                                          |
| <b>Description</b>           | Allow an authorized user(admin, students, parents and teachers) to securely log in to the system to access their respective functionalities.                                   |
| <b>Primary Actors</b>        | Admin, Parent, Teacher, Student                                                                                                                                                |
| <b>Preconditions</b>         | The user has a valid username and password.<br>The system is operational and the login page is displayed.                                                                      |
| <b>Post conditions</b>       | The user is successfully authenticated and redirected to the admin dashboard.<br>An error message is displayed for invalid credentials and the user remains on the login page. |
| <b>Main success scenario</b> | User open the app<br>User enter username/email and password<br>System verify credentials<br>User is logged in successfully                                                     |
| <b>Alternate Flow</b>        | User enter incorrect credentials<br>System display an error message                                                                                                            |
| <b>Exception</b>             | Server error or network failure during login                                                                                                                                   |
| <b>Status</b>                | Draft                                                                                                                                                                          |
| <b>Priority</b>              | Medium                                                                                                                                                                         |
| <b>Include</b>               | User authentication Role based access control                                                                                                                                  |



*Table 8 Use Case 4: Update Profile*

|                              |                                                                                                                                                                                                      |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b>                   | UC#4                                                                                                                                                                                                 |
| <b>Title</b>                 | Update Profile                                                                                                                                                                                       |
| <b>Description</b>           | Allow an authorized user(admin, students, parents and teachers) to modify their personal information such as contact details, username etc.                                                          |
| <b>Primary Actors</b>        | Admin, Parent, Teacher, Student                                                                                                                                                                      |
| <b>Preconditions</b>         | The user is successfully logged into the system.<br>The user is on the “Profile” or “Settings” page.                                                                                                 |
| <b>Post conditions</b>       | The user’s information is updated in the database.<br>A confirmation message “Profile updated successfully” is displayed.                                                                            |
| <b>Main success scenario</b> | User selects “Edit Profile”.<br>User modifies the desired fields.<br>User clicks the “Update” button.<br>System validates the input data.<br>System updates the record and displays success message. |
| <b>Alternate Flow</b>        | User enters data in an incorrect format.<br>System displays a validation error.                                                                                                                      |
| <b>Exception</b>             | Database connection error, server timeout during update.                                                                                                                                             |
| <b>Status</b>                | Draft                                                                                                                                                                                                |
| <b>Priority</b>              | Medium                                                                                                                                                                                               |
| <b>Include</b>               | Personal details, Contact info                                                                                                                                                                       |





*Table 9 Use Case 5: Approve student and parent registration*

|                              |                                                                                                                                                                                                                   |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b>                   | UC#5                                                                                                                                                                                                              |
| <b>Title</b>                 | Approve student and parent registrations                                                                                                                                                                          |
| <b>Description</b>           | Allows an administrator to review and approve registration requests from students and parents                                                                                                                     |
| <b>Primary Actors</b>        | Admin                                                                                                                                                                                                             |
| <b>Preconditions</b>         | Student or parent has submitted a registration                                                                                                                                                                    |
| <b>Post conditions</b>       | The users (student and parent) registration is approved and their account is activated. Generate a unique ID of Student. A notification is sent to the approved user. The request is remove from the pending list |
| <b>Main success scenario</b> | Admin log in.<br>Admin view pending registration.<br>Admin select a request.<br>Admin approve the registration.<br>System activate the account and generate a unique key for Student                              |
| <b>Alternate Flow</b>        | Admin reject the registration request.<br>System send rejection notification.                                                                                                                                     |
| <b>Exception</b>             | System error or connection failure during approval process.                                                                                                                                                       |
| <b>Status</b>                | Draft                                                                                                                                                                                                             |
| <b>Priority</b>              | Medium                                                                                                                                                                                                            |
| <b>Include</b>               | User authentication Notification Service                                                                                                                                                                          |



*Table 10 Use Case 6: Add, update and delete students, teachers and parents*

|                              |                                                                                                                                                                                                    |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b>                   | UC#6                                                                                                                                                                                               |
| <b>Title</b>                 | Add, update and delete students, teachers and parents                                                                                                                                              |
| <b>Description</b>           | Allow the admin to add new users, update existing users details or delete student, teacher and parent accounts.                                                                                    |
| <b>Primary Actors</b>        | Admin                                                                                                                                                                                              |
| <b>Preconditions</b>         | Admin is logged in and has management privileges.                                                                                                                                                  |
| <b>Post conditions</b>       | User records are added, updated or removed from the system database.                                                                                                                               |
| <b>Main success scenario</b> | Admin log in<br>Admin select the users(Students, teachers and Parents) .<br>Admin choose add, update or delete option.<br>Admin enter or modify user details.<br>System save changes successfully. |
| <b>Alternate flow</b>        | Admin cancel the operation.<br>No changes are saved.                                                                                                                                               |
| <b>Exception</b>             | Database error or system failure during operation.                                                                                                                                                 |
| <b>Status</b>                | Approved                                                                                                                                                                                           |
| <b>Priority</b>              | High                                                                                                                                                                                               |
| <b>Include</b>               | User Authentication Data Validation                                                                                                                                                                |



*Table 11 Use Case 7: Manage Announcements*

|                              |                                                                                                                              |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b>                   | UC#7                                                                                                                         |
| <b>Title</b>                 | Manage announcement                                                                                                          |
| <b>Description</b>           | Admin can create, edit and delete system announcement for students, teachers and parents.                                    |
| <b>Primary Actors</b>        | Admin                                                                                                                        |
| <b>Preconditions</b>         | Admin must be logged in and have permissions to manage announcements.                                                        |
| <b>Post conditions</b>       | Announcements are updated and relevant users can view them via notifications.                                                |
| <b>Main success scenario</b> | Admin log in.<br>Admin open Announcement module.<br>Admin create or edit an announcement.<br>Admin publish the announcement. |
| <b>Alternate flow</b>        | Admin deletes an announcement.<br>System removes it from user view.                                                          |
| <b>Exception</b>             | System or database error during announcement management.                                                                     |
| <b>Status</b>                | Approved                                                                                                                     |
| <b>Priority</b>              | High                                                                                                                         |
| <b>Include</b>               | User Authentication Notification Service                                                                                     |

*Table 12 Use Case 8: View system reports*

|                              |                                                                                                                     |
|------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b>                   | UC#8                                                                                                                |
| <b>Title</b>                 | View system report                                                                                                  |
| <b>Description</b>           | Admin can view various system report in the system.                                                                 |
| <b>Primary Actors</b>        | Admin                                                                                                               |
| <b>Preconditions</b>         | Admin must be logged in.                                                                                            |
| <b>Post conditions</b>       | The system displays the requested report.                                                                           |
| <b>Main success scenario</b> | Admin log in.<br>Admin select System Report.<br>Admin choose the report.<br>System generate and display the report. |
| <b>Alternate flow</b>        | Admin applies filters.<br>System update the report view.                                                            |
| <b>Exception</b>             | Report generation failure or database error.                                                                        |
| <b>Status</b>                | Approved                                                                                                            |
| <b>Priority</b>              | Medium                                                                                                              |
| <b>Include</b>               | User Authentication Reporting Module                                                                                |

*Table 13 Use Case 9: Record Fee Payment*

|                              |                                                                                                                          |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b>                   | UC#9                                                                                                                     |
| <b>Title</b>                 | Record fee payment                                                                                                       |
| <b>Description</b>           | Admin can save fee payment record from a student in the system.                                                          |
| <b>Primary Actors</b>        | Admin                                                                                                                    |
| <b>Preconditions</b>         | Admin must be logged in and the student record must exist.                                                               |
| <b>Post conditions</b>       | The fee payment is recorded in the system and fee paid notification send to student and parents.                         |
| <b>Main success scenario</b> | Admin log in.<br>Admin select.<br>Admin search student record.<br>Admin enter payment detail.<br>System save fee record. |
| <b>Alternate flow</b>        | Admin update an existing payment record.<br>System refresh fee status.                                                   |
| <b>Exception</b>             | Payment record save failure or database error.                                                                           |
| <b>Status</b>                | Approved                                                                                                                 |
| <b>Priority</b>              | High                                                                                                                     |
| <b>Include</b>               | User Authentication Fee Management Module                                                                                |



*Table 14 Use Case10: Classes timetable*

|                              |                                                                                                                                                                      |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b>                   | UC#10                                                                                                                                                                |
| <b>Title</b>                 | Classes timetable                                                                                                                                                    |
| <b>Description</b>           | Allow an admin to create and modify weekly class schedule within the system and allow the student and teacher to view the classes timetable.                         |
| <b>Primary Actors</b>        | Admin                                                                                                                                                                |
| <b>Preconditions</b>         | The admin must be logged into the application and the admin have appropriate permission to edit scheduled and teacher and student have permission to view timetable. |
| <b>Post conditions</b>       | The new timetable is saved and published to the system visible to the relevant user student and teacher name.                                                        |
| <b>Main success scenario</b> | User log in to the system.<br>User open timetable section.<br>System retrieves timetable data.<br>System display class timetable.                                    |
| <b>Alternate flow</b>        | Admin change day or view weekly timetable.                                                                                                                           |
| <b>Exception</b>             | Timetable not available or system error                                                                                                                              |
| <b>Status</b>                | Approved                                                                                                                                                             |
| <b>Priority</b>              | Medium                                                                                                                                                               |
| <b>Include</b>               | Login                                                                                                                                                                |

*Table 15 Use Case11: View enrolled subjects*

|                              |                                                                                                                     |
|------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b>                   | UC#11                                                                                                               |
| <b>Title</b>                 | View enrolled subjects                                                                                              |
| <b>Description</b>           | Students view enrolled subjects                                                                                     |
| <b>Primary Actors</b>        | Student                                                                                                             |
| <b>Preconditions</b>         | The student should be logged in.                                                                                    |
| <b>Post conditions</b>       | Enrolled subjects displayed on screen.                                                                              |
| <b>Main success scenario</b> | Student log in.<br>Student select the subjects.<br>System retrieve enrolled subjects.<br>Subject list is displayed. |
| <b>Alternate flow</b>        | No subject enrolled.<br>System display a message.                                                                   |
| <b>Exception</b>             | Data retrieval failure or system error                                                                              |
| <b>Status</b>                | Approved                                                                                                            |
| <b>Priority</b>              | Medium                                                                                                              |
| <b>Include</b>               | User Authentication Subject Management Module                                                                       |



*Table 16 Use Case12: Attend online classes*

|                              |                                                                                                                                                                                                                                  |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b>                   | UC#12                                                                                                                                                                                                                            |
| <b>Title</b>                 | Attend live classes                                                                                                                                                                                                              |
| <b>Description</b>           | Student joins and attend the scheduled online class using the provided link.                                                                                                                                                     |
| <b>Primary Actors</b>        | Student                                                                                                                                                                                                                          |
| <b>Preconditions</b>         | The student is registered and must be logged into the application.                                                                                                                                                               |
| <b>Post conditions</b>       | Student successfully attends the online class.                                                                                                                                                                                   |
| <b>Main success scenario</b> | <p>Student log in.</p> <p>Student view online class schedule.</p> <p>Student clicks on the class meeting link.</p> <p>System redirects to the online meeting platform.</p> <p>Student joins the meeting and attend the class</p> |
| <b>Alternate flow</b>        | <p>Student join the class directly from the notification link.</p> <p>Student joins the class late.</p>                                                                                                                          |
| <b>Exception</b>             | <p>Invalid or expire link shows an error message.</p> <p>Internet or system issue unable to join the class.</p>                                                                                                                  |
| <b>Status</b>                | Approved                                                                                                                                                                                                                         |
| <b>Priority</b>              | High                                                                                                                                                                                                                             |
| <b>Include</b>               | Authenticate student view class schedule                                                                                                                                                                                         |





*Table 17 Use Case13: Download Study Material*

|                              |                                                                                                            |
|------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>ID:</b>                   | UC#13                                                                                                      |
| <b>Title</b>                 | Download study material                                                                                    |
| <b>Description</b>           | Registered students can download available study materials from the app.                                   |
| <b>Primary Actors</b>        | Student                                                                                                    |
| <b>Preconditions</b>         | The student must be logged in and study material must be available.                                        |
| <b>Post conditions</b>       | The selected study material is downloaded to the student's device.                                         |
| <b>Main success scenario</b> | Student log in.<br>Student select subject.<br>Student choose study material.<br>System downloads the file. |
| <b>Alternate flow</b>        | Student view material online instead of downloading.<br>System displays the content.                       |
| <b>Exception</b>             | Download failure or network issue.                                                                         |
| <b>Status</b>                | Proposed                                                                                                   |
| <b>Priority</b>              | High                                                                                                       |
| <b>Include</b>               | User Authentication File Download                                                                          |



*Table 18 Use Case14: View and attempt online quizzes and assignment*

|                              |                                                                                                                                                                  |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b>                   | UC#14                                                                                                                                                            |
| <b>Title</b>                 | View and attempt quizzes and assignment                                                                                                                          |
| <b>Description</b>           | Allow a student to view an available quiz and assignments within the app.                                                                                        |
| <b>Primary Actors</b>        | Student                                                                                                                                                          |
| <b>Preconditions</b>         | The student must be registered and logged into the application.                                                                                                  |
| <b>Post conditions</b>       | The student must be logged into the system and a quiz and assignment must be available and assigned to them.                                                     |
| <b>Main success scenario</b> | Student log in.<br>Student select Quizzes and assignment.<br>Student select an available quiz and assignment.<br>Student attempt and submits quiz and assignment |
| <b>Alternate flow</b>        | Student exits quiz and assignment before submission.<br>System auto save or discard attempt.                                                                     |
| <b>Exception</b>             | Timer expiration or network failure during quiz and assignment.<br>.                                                                                             |
| <b>Status</b>                | Proposed                                                                                                                                                         |
| <b>Priority</b>              | High                                                                                                                                                             |
| <b>Include</b>               | User Authentication Quiz and assignment                                                                                                                          |



*Table 19 Use Case 15: View fee history and status*

|                              |                                                                                                                                                                                                                                                                                                                                                |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b>                   | UC#15                                                                                                                                                                                                                                                                                                                                          |
| <b>Title</b>                 | View fee history and status                                                                                                                                                                                                                                                                                                                    |
| <b>Description</b>           | Allow parent to view the fee history and payment status of a linked child's account.                                                                                                                                                                                                                                                           |
| <b>Primary Actors</b>        | Parent, student                                                                                                                                                                                                                                                                                                                                |
| <b>Preconditions</b>         | The parent and the child must have an active account, the parent account must be linked with the child's account and the fee data must exist for the child.                                                                                                                                                                                    |
| <b>Post conditions</b>       | The child's fee history and status are successfully displayed on the parent's dashboard.                                                                                                                                                                                                                                                       |
| <b>Main success scenario</b> | <p>Parent/Student logs into the system.</p> <p>Selects "View Fee History and Status" option.</p> <p>System displays detailed payment history, including dates, amounts and status (paid/unpaid).</p> <p>Parent/Student reviews current balance and pending payments.</p> <p>Optionally, Parent/Student downloads or prints the fee report.</p> |
| <b>Alternate flow</b>        | <p>Parent/Student filters fee history by month or year.</p> <p>Parent/Student receives a summary of total fees paid and pending.</p>                                                                                                                                                                                                           |
| <b>Exception</b>             | <p>No fee records found for the student.</p> <p>System error prevents displaying the fee history.</p>                                                                                                                                                                                                                                          |
| <b>Status</b>                | Approved                                                                                                                                                                                                                                                                                                                                       |
| <b>Priority</b>              | High                                                                                                                                                                                                                                                                                                                                           |
| <b>Include</b>               | Login, Link parent account to child, notification for fee payment                                                                                                                                                                                                                                                                              |



*Table 20 Use Case 16: Notification for Fee payment*

|                              |                                                                                                                                                                                                                                                                                    |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b>                   | UC#16                                                                                                                                                                                                                                                                              |
| <b>Title</b>                 | Notification for Fee payment                                                                                                                                                                                                                                                       |
| <b>Description</b>           | Allow parent to pay their child's fees through the system.                                                                                                                                                                                                                         |
| <b>Primary Actors</b>        | Parent, student                                                                                                                                                                                                                                                                    |
| <b>Preconditions</b>         | Automatically notifies parents or students about pending or upcoming fee payments, including due dates and payment status.                                                                                                                                                         |
| <b>Post conditions</b>       | Parent or student receives a notification about pending or due fees.                                                                                                                                                                                                               |
| <b>Main success scenario</b> | <p>System checks pending or upcoming fee payments.</p> <p>System generates a notification.</p> <p>System sends notification to parent/student via app, email, or SMS.</p> <p>Parent/student receives the notification.</p> <p>Parent/student views payment details and status.</p> |
| <b>Alternate flow</b>        | <p>Parent/student has multiple children; system sends consolidated notifications.</p> <p>Parent/student views history of previous fee notifications.</p>                                                                                                                           |
| <b>Exception</b>             | <p>Fee record not found for the student.</p> <p>Notification delivery fails due to network or system error.</p>                                                                                                                                                                    |
| <b>Status</b>                | Approved                                                                                                                                                                                                                                                                           |
| <b>Priority</b>              | High                                                                                                                                                                                                                                                                               |
| <b>Include</b>               | Login, Link parent account to child, view fee history                                                                                                                                                                                                                              |



*Table 21 Use Case17: Create link for online classes*

|                              |                                                                                                                                                                                                                                                       |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b>                   | UC#17                                                                                                                                                                                                                                                 |
| <b>Title</b>                 | Create link for online classes                                                                                                                                                                                                                        |
| <b>Description</b>           | Teacher creates and shares an online class link with the students.                                                                                                                                                                                    |
| <b>Primary Actors</b>        | Teacher                                                                                                                                                                                                                                               |
| <b>Preconditions</b>         | The teacher must be logged into the application.                                                                                                                                                                                                      |
| <b>Post conditions</b>       | Online classes link is created and available to the students.                                                                                                                                                                                         |
| <b>Main success scenario</b> | <p>Teacher log in.</p> <p>Teacher select create online Class.</p> <p>Teacher enter class detail (subject, date, time).</p> <p>Teacher enters meeting link.</p> <p>Teacher click saves/ share.</p> <p>System saves the link and notifies students.</p> |
| <b>Alternate flow</b>        | <p>Teacher edit class detail.</p> <p>Teacher reschedule the class.</p>                                                                                                                                                                                |
| <b>Exception</b>             | <p>Invalid or missing meeting link error shown.</p> <p>System or internet failure.</p>                                                                                                                                                                |
| <b>Status</b>                | Approved                                                                                                                                                                                                                                              |
| <b>Priority</b>              | High                                                                                                                                                                                                                                                  |
| <b>Include</b>               | Authenticate teacher notify students                                                                                                                                                                                                                  |

*Table 22 Use Case18: Upload study material*

|                              |                                                                                                                                       |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b>                   | UC#18                                                                                                                                 |
| <b>Title</b>                 | Upload study material                                                                                                                 |
| <b>Description</b>           | Allow a teacher to upload study material for students within the app.                                                                 |
| <b>Primary Actors</b>        | Teacher                                                                                                                               |
| <b>Preconditions</b>         | The teacher must be logged into the application. The teacher must have a course assigned to them.                                     |
| <b>Post conditions</b>       | Study material must be available for students to view.                                                                                |
| <b>Main success scenario</b> | Teachers log in.<br>Teacher select Study Materials.<br>Teacher upload file and add details.<br>System validate and save the material. |
| <b>Alternate flow</b>        | Teacher update or delete existing material.<br>System reflect changes.                                                                |
| <b>Exception</b>             | File upload failure or unsupported file format                                                                                        |
| <b>Status</b>                | Approved                                                                                                                              |
| <b>Priority</b>              | High                                                                                                                                  |
| <b>Include</b>               | User Authentication File Management                                                                                                   |



*Table 23 Use Case19: Upload and view assignments*

|                              |                                                                                                                                        |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b>                   | UC#19                                                                                                                                  |
| <b>Title</b>                 | Upload and view assignments                                                                                                            |
| <b>Description</b>           | Teachers upload assignment and view assignment for their classes.                                                                      |
| <b>Primary Actors</b>        | Teacher                                                                                                                                |
| <b>Preconditions</b>         | The teacher must be logged in.                                                                                                         |
| <b>Post conditions</b>       | Assignment details displayed on screen or confirmation of upload received.                                                             |
| <b>Main success scenario</b> | Teacher select subject.<br>Teacher click upload assignment.<br>Teacher open assignment section.<br>System display uploaded assignments |
| <b>Alternate flow</b>        | Teacher edit, update or reload the assignment.                                                                                         |
| <b>Exception</b>             | Invalid file format, upload failure or assignment not found                                                                            |
| <b>Status</b>                | Approved                                                                                                                               |
| <b>Priority</b>              | Medium                                                                                                                                 |
| <b>Include</b>               | Log in                                                                                                                                 |



*Table 24 Use Case20: Create online quizzes*

|                              |                                                                                                                                                            |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b>                   | UC#20                                                                                                                                                      |
| <b>Title</b>                 | Create quizzes                                                                                                                                             |
| <b>Description</b>           | Allow a teacher to create quizzes within the system.                                                                                                       |
| <b>Primary Actors</b>        | Teacher                                                                                                                                                    |
| <b>Preconditions</b>         | The teacher must be logged into the application. The teacher must have a course assigned to them.                                                          |
| <b>Post conditions</b>       | Study material must be available for students to view.                                                                                                     |
| <b>Main success scenario</b> | Teacher log in.<br>Teacher select Quizzes.<br>Teacher create quiz questions and settings.<br>Teacher will set the quiz timer.<br>Teacher publish the quiz. |
| <b>Alternate flow</b>        | Teacher edit or delete a quiz.<br>System update quiz details.                                                                                              |
| <b>Exception</b>             | Quiz save failure or system error.                                                                                                                         |
| <b>Status</b>                | Approved                                                                                                                                                   |
| <b>Priority</b>              | High                                                                                                                                                       |
| <b>Include</b>               | User Authentication Quiz Management                                                                                                                        |



*Table 25 Use Case 21: Manage student performance*

|                              |                                                                                                                                                                                                      |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b>                   | UC#21                                                                                                                                                                                                |
| <b>Title</b>                 | Manage student performance                                                                                                                                                                           |
| <b>Description</b>           | Allow teacher to record, update and view student grades and performance metrics.                                                                                                                     |
| <b>Primary Actors</b>        | Teacher                                                                                                                                                                                              |
| <b>Preconditions</b>         | The teacher must be logged in and assigned to relevant courses.                                                                                                                                      |
| <b>Post conditions</b>       | Student performance data is recorded and updated in the system.                                                                                                                                      |
| <b>Main success scenario</b> | Teacher log in to the system.<br>Teacher select class or subject.<br>Teacher view student performance list.<br>Teacher enter or update marks/grades.<br>System save and display updated performance. |
| <b>Alternate flow</b>        | Teacher edit or update previously entered performance data.                                                                                                                                          |
| <b>Exception</b>             | Invalid data entry.<br>Student record not found or system error.                                                                                                                                     |
| <b>Status</b>                | Approved                                                                                                                                                                                             |
| <b>Priority</b>              | High                                                                                                                                                                                                 |
| <b>Include</b>               | Login                                                                                                                                                                                                |

*Table 26 Use Case22: Communication with Parents*

|                              |                                                                                                                                                                                                                                           |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b>                   | UC#22                                                                                                                                                                                                                                     |
| <b>Title</b>                 | Communication with Parents                                                                                                                                                                                                                |
| <b>Description</b>           | System allow teachers to send messages or update to parents and parents to view and respond to these communication.                                                                                                                       |
| <b>Primary Actors</b>        | Teacher, Parent                                                                                                                                                                                                                           |
| <b>Preconditions</b>         | The teacher or parents must be logged in.                                                                                                                                                                                                 |
| <b>Post conditions</b>       | Message or notification is delivered to parents.                                                                                                                                                                                          |
| <b>Main success scenario</b> | Teacher logs in to the system.<br>Teacher selects student or parent<br>Teacher write a message or announcements.<br>Teacher sends a message.<br>System delivers the message to the parent through the email.<br>Parent views the message. |
| <b>Alternate flow</b>        | Teacher edits or resends the message before final submission.<br>Parent views the message later if not online.                                                                                                                            |
| <b>Exception</b>             | Parent account not found.<br>Message sending fails due to system error.                                                                                                                                                                   |
| <b>Status</b>                | Approved                                                                                                                                                                                                                                  |
| <b>Priority</b>              | High                                                                                                                                                                                                                                      |
| <b>Include</b>               | Login                                                                                                                                                                                                                                     |



*Table 27 Use Case23: Link parent account to child*

|                              |                                                                                                                                                                                                                                                                                            |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b>                   | UC#23                                                                                                                                                                                                                                                                                      |
| <b>Title</b>                 | Link child to parent                                                                                                                                                                                                                                                                       |
| <b>Description</b>           | Allow child to link their account to a parent existing account.                                                                                                                                                                                                                            |
| <b>Primary Actors</b>        | Parent                                                                                                                                                                                                                                                                                     |
| <b>Preconditions</b>         | The parent and the child must have an active account.                                                                                                                                                                                                                                      |
| <b>Post conditions</b>       | The child account is successfully linked to the parent account.                                                                                                                                                                                                                            |
| <b>Main success scenario</b> | Parent logs in to the system.<br>Parent selects “Link child account” option.<br>Parent enters child unique ID or registration number.<br>System verifies the child account.<br>System links the child account to the parent account.<br>Child receives confirmation of successful linking. |
| <b>Alternate flow</b>        | Parent requests linking for multiple children at once.<br>Parent updates or removes an existing linked child account.                                                                                                                                                                      |
| <b>Exception</b>             | Child account not found.<br>Parent provide incorrect child ID.<br>System error prevents linking.                                                                                                                                                                                           |
| <b>Status</b>                | Approved                                                                                                                                                                                                                                                                                   |
| <b>Priority</b>              | High                                                                                                                                                                                                                                                                                       |
| <b>Include</b>               | Login                                                                                                                                                                                                                                                                                      |



*Table 28 Use Case24: View child profile*

|                              |                                                                                                                                                                                                                                                       |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b>                   | UC#24                                                                                                                                                                                                                                                 |
| <b>Title</b>                 | View child profile                                                                                                                                                                                                                                    |
| <b>Description</b>           | Allow parent to view the profile information of a linked child's account.                                                                                                                                                                             |
| <b>Primary Actors</b>        | Parent                                                                                                                                                                                                                                                |
| <b>Preconditions</b>         | The parent and the child must have an active account and the parent account must be linked with the child's account.                                                                                                                                  |
| <b>Post conditions</b>       | The child's profile detail is successfully displayed to the parent.                                                                                                                                                                                   |
| <b>Main success scenario</b> | Parent logs into the system.<br>Parent selects "View Child Profile" option.<br>System displays the child's personal details.<br>System displays academic information, attendance records, and performance metrics.<br>Parent reviews the information. |
| <b>Alternate flow</b>        | Parent navigates between multiple children if more than one child is linked.<br>Parent updates notification preferences while viewing profile.                                                                                                        |
| <b>Exception</b>             | No child account linked to parent.<br>System error prevents loading the profile.                                                                                                                                                                      |
| <b>Status</b>                | Approved                                                                                                                                                                                                                                              |
| <b>Priority</b>              | High                                                                                                                                                                                                                                                  |
| <b>Include</b>               | Login, Link parent account to child                                                                                                                                                                                                                   |



Table 29 Use Case 25: View result

|                              |                                                                                                                                                                                                                                                                                                 |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b>                   | UC#25                                                                                                                                                                                                                                                                                           |
| <b>Title</b>                 | View result                                                                                                                                                                                                                                                                                     |
| <b>Description</b>           | Allow parent and students to view the academic result.                                                                                                                                                                                                                                          |
| <b>Primary Actors</b>        | Parent, Students                                                                                                                                                                                                                                                                                |
| <b>Preconditions</b>         | The parent and the child must have an active account. The parent account must be linked with the child's account and the child must have result available in the system.                                                                                                                        |
| <b>Post conditions</b>       | The child's academic results are successfully displayed to the parent and student dashboard.                                                                                                                                                                                                    |
| <b>Main success scenario</b> | <p>Parent/Student logs into the system.</p> <p>Selects "View Results" option.</p> <p>System displays the student's exam results and grades.</p> <p>System shows performance trends, total marks and subject-wise marks.</p> <p>Parent/Student reviews and optionally downloads the results.</p> |
| <b>Alternate flow</b>        | <p>Parent/Student selects a specific class to view results.</p> <p>System provides graphical representation of performance.</p>                                                                                                                                                                 |
| <b>Exception</b>             | <p>Results not yet available.</p> <p>Parent/Student account not linked correctly.</p> <p>System error prevents loading results.</p>                                                                                                                                                             |
| <b>Status</b>                | Approved                                                                                                                                                                                                                                                                                        |
| <b>Priority</b>              | High                                                                                                                                                                                                                                                                                            |
| <b>Include</b>               | Login, Link child account to parent                                                                                                                                                                                                                                                             |



*Table 30 Use Case26: Receive announcements*

|                              |                                                                                                                                                    |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b>                   | UC#26                                                                                                                                              |
| <b>Title</b>                 | Receive announcement                                                                                                                               |
| <b>Description</b>           | Allows users (Student, Teachers, Parents) to view available announcements within the app through notification.                                     |
| <b>Primary Actors</b>        | Student, Teachers, Parents                                                                                                                         |
| <b>Preconditions</b>         | The user (Student, Teachers, Parents) must be logged into the application.                                                                         |
| <b>Post conditions</b>       | Announcement must be available and assigned to them.                                                                                               |
| <b>Main success scenario</b> | User logs in to the system.<br>User open the announcement section.<br>System retrieves announcements.<br>System display announcements to the user. |
| <b>Alternate flow</b>        | User refresh or filter announcement                                                                                                                |
| <b>Exception</b>             | No announcement or system error                                                                                                                    |
| <b>Status</b>                | Approved                                                                                                                                           |
| <b>Priority</b>              | Medium                                                                                                                                             |
| <b>Include</b>               | Login                                                                                                                                              |

*Table31 Use Case 27: Feedback and Review*

|                              |                                                                        |
|------------------------------|------------------------------------------------------------------------|
| <b>Id</b>                    | UC#27                                                                  |
| <b>Title</b>                 | Feedback and Review                                                    |
| <b>Description</b>           | Allows student and parent to give feedback and review about the system |
| <b>Actor</b>                 | Parent, student                                                        |
| <b>Preconditions</b>         | Parent and student must enter feedback                                 |
| <b>Post conditions</b>       | Parent and student can give feedback and reviews of using app.         |
| <b>Main success scenario</b> | Student and parent give review about system(App).                      |
| <b>Alternate Flow</b>        | None                                                                   |
| <b>Exceptions</b>            | Duplicate Review<br>Submission Failure                                 |
| <b>Status</b>                | Completed                                                              |
| <b>Priority</b>              | Medium                                                                 |
| <b>Include</b>               | None                                                                   |



*Table 32 Use Case 28: Logout*

|                              |                                                                                                                                                                             |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b>                   | UC#28                                                                                                                                                                       |
| <b>Title</b>                 | Logout                                                                                                                                                                      |
| <b>Description</b>           | The user ends their session and logout of the system.                                                                                                                       |
| <b>Primary Actors</b>        | Admin, Students, Teachers, Parents                                                                                                                                          |
| <b>Preconditions</b>         | The user is currently logged in.                                                                                                                                            |
| <b>Post conditions</b>       | The user is logged out and the system displays the login screen.                                                                                                            |
| <b>Main success scenario</b> | User clicks the “Logout” button.<br>System terminates the user session.<br>System redirects the user to the login page.<br>User confirms they have successfully logged out. |
| <b>Alternate flow</b>        | User session times out automatically due to inactivity, triggering automatic logout.<br>User cancels logout confirmation (if system asks for confirmation).                 |
| <b>Exception</b>             | System error prevents logout (rare).<br>Network interruption during logout may require user to retry.                                                                       |
| <b>Status</b>                | Approved                                                                                                                                                                    |
| <b>Priority</b>              | High                                                                                                                                                                        |
| <b>Include</b>               | Login                                                                                                                                                                       |





## Storyboarding

This story board is made using Canva. It depicts our student module in visual form.

# KIPS COACHING KHARIAN



*Figure 3: Story boarding*

## 3.3 Functional Requirements

Requirements that must be fulfilled to complete the functionality of our system.

- 1.User Registration:** User (student and parent) shall be able to register themselves by providing credentials.
- 2.Reset Password:** User can reset password in the case of forgot password.
- 3.User Login:** User shall be able to login to the system after registration.



- 4.Update Profile:** Users (Admin, student, teacher and parent) shall update their profiles.
- 5.Approve student and parent registration:** Admin shall approve the student and parent registration and manually add the teacher.
- 6.Add, update and delete users:** Admin shall add, update and delete users (student, teacher and parents).
- 7.Manage announcements:** Admin shall manage the announcements for the students, teachers and parents.
- 8.View System Report:** Admin shall view the system report.
- 9.Record fee payment:** Admin shall record fee payment in the system.
- 10.View enrolled subjects:** Student shall view the enrolled subjects.
- 11.Attend live classes:** Students shall attend live classes
- 12.Download study material:** Students shall download the study material.
- 13.View and attempt quizzes and assignment:** Students shall view and attempt quizzes and assignments.
- 14.Upload Study material:** Teacher shall upload the study material.
- 15.Create quizzes:** Teacher shall create quizzes for the students.
- 16.Upload and view assignments:** Teacher shall upload and view assignments.
- 17.Create link for online classes:** Teacher shall create a link for online class.
- 18.Receive announcement:** Users (students, teachers and parents) shall receive announcements.
- 19.Classes timetable:** Admin shall provide the timetable and the teachers and students shall view the classes timetable.
- 20.Manage student performance:** Teacher shall manage the student performance.
- 21.Communication with parents:** Teacher shall communicate with the parents through the email.
- 22.Link parent account to child:** The parent account shall link with the child.
- 23. View child profile:** Parents shall view the profile information of the linked child's account.
- 24.View result:** The students and parents shall view the results.
- 25.Notification for fee payment:** Allow parents and students to pay their fee.
- 26.View fee history and status:** The parent shall view the fee history and payment status of linked child's account.
- 27. Feedback and Review:** Students and Parents give the review about the app performance.
- 28.Logout:** The user ends their session and logout of the system.



### 3.3.1 Functional Requirements

Table 33 Functional Requirement#1

|                                    |                                                                          |
|------------------------------------|--------------------------------------------------------------------------|
| <b>Identifier</b>                  | Functional Requirement#1                                                 |
| <b>Title</b>                       | Register                                                                 |
| <b>Requirement</b>                 | User can register for an account using a unique ID and personal details. |
| <b>Source</b>                      | KIPS management                                                          |
| <b>Rationale</b>                   | To allow users to access the system and its features.                    |
| <b>Business Rule (if required)</b> | User must have a valid ID.                                               |
| <b>Dependencies</b>                | None                                                                     |
| <b>Priority</b>                    | High                                                                     |

Table 34 Functional Requirement#2

|                                    |                                                                                                                                                             |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Identifier</b>                  | Functional Requirement#2                                                                                                                                    |
| <b>Title</b>                       | Reset password                                                                                                                                              |
| <b>Requirement</b>                 | System allows users to securely reset their forgotten password.                                                                                             |
| <b>Source</b>                      | System requirement                                                                                                                                          |
| <b>Rationale</b>                   | To allow users to regain access to the system if they forget their credentials.                                                                             |
| <b>Business Rule (if required)</b> | User must provide a valid email address or username to receive a reset link or code. The reset token should expire within a short period and be single use. |
| <b>Dependencies</b>                | User authentication system, email service.                                                                                                                  |



|                 |      |
|-----------------|------|
| <b>Priority</b> | High |
|-----------------|------|

Table 35 Functional Requirement#3

|                                    |                                                                                    |
|------------------------------------|------------------------------------------------------------------------------------|
| <b>Identifier</b>                  | Functional Requirement#3                                                           |
| <b>Title</b>                       | Log in                                                                             |
| <b>Requirement</b>                 | System allows users to securely login into the system                              |
| <b>Source</b>                      | System requirement                                                                 |
| <b>Rationale</b>                   | To provide secure access for system users.                                         |
| <b>Business Rule (if required)</b> | User must be an admin, student, teacher or parent and login with valid credentials |
| <b>Dependencies</b>                | User login system user authentication                                              |
| <b>Priority</b>                    | High                                                                               |

Table 36 Functional Requirement#4

|                                    |                                                                                                                      |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Identifier</b>                  | Functional Requirement#4                                                                                             |
| <b>Title</b>                       | Update profile                                                                                                       |
| <b>Requirement</b>                 | User shall be able to view and update their personal information like username, email, password and profile picture. |
| <b>Source</b>                      | User, Security standards                                                                                             |
| <b>Rationale</b>                   | Allow user to keep their account information accurate and secure.                                                    |
| <b>Business Rule (if required)</b> | User must be registered.<br>Password must meet requirements                                                          |
| <b>Dependencies</b>                | FR-2                                                                                                                 |
| <b>Priority</b>                    | Medium                                                                                                               |

Table 37 Functional Requirement#5



|                                    |                                                                                                                             |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Identifier</b>                  | Functional Requirement#5                                                                                                    |
| <b>Title</b>                       | Approve student and parent registration                                                                                     |
| <b>Requirement</b>                 | System must verify the student and parent identity during the registration process to ensure authenticity and prevent fraud |
| <b>Source</b>                      | System design specification legal compliance                                                                                |
| <b>Rationale</b>                   | To maintain data integrity security and comply with regulations                                                             |
| <b>Business Rule (if required)</b> | User must provide a valid e-mail address and a unique identification document                                               |
| <b>Dependencies</b>                | E-mail system, ID verification services or database, user database                                                          |
| <b>Priority</b>                    | High                                                                                                                        |

Table 38 Functional Requirement#6

|                                    |                                                                                                  |
|------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Identifier</b>                  | Functional Requirement#6                                                                         |
| <b>Title</b>                       | Add, update and delete users                                                                     |
| <b>Requirement</b>                 | Allows an admin to securely add a new users account to the system                                |
| <b>Source</b>                      | kips requirements and parent feedback                                                            |
| <b>Rationale</b>                   | To efficiently enroll new users and manage records                                               |
| <b>Business Rule (if required)</b> | User must be logged in and have appropriate permissions and requirement fields must be completed |
| <b>Dependencies</b>                | User login, user database, account creation system                                               |
| <b>Priority</b>                    | High                                                                                             |

Table 39 Functional Requirement#7

|                    |                                                                                   |
|--------------------|-----------------------------------------------------------------------------------|
| <b>Identifier</b>  | Functional Requirement#7                                                          |
| <b>Title</b>       | Manage announcement                                                               |
| <b>Requirement</b> | The system shall allow administrators to create, update and delete announcements. |



|                                    |                                                                                               |
|------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Source</b>                      | System user                                                                                   |
| <b>Rationale</b>                   | To communicate important information to users effectively and maintain current announcements. |
| <b>Business Rule (if required)</b> | Only administrators can manage announcements.                                                 |
| <b>Dependencies</b>                | Database connection                                                                           |
| <b>Priority</b>                    | High                                                                                          |

Table 40 Functional Requirement#8

|                                    |                                                               |
|------------------------------------|---------------------------------------------------------------|
| <b>Identifier</b>                  | Functional Requirement#8                                      |
| <b>Title</b>                       | View system reports                                           |
| <b>Requirement</b>                 | The system shall allow administrators to view system reports. |
| <b>Source</b>                      | System user                                                   |
| <b>Rationale</b>                   | To monitor system performance and usage.                      |
| <b>Business Rule (if required)</b> | Only administrator can view the report.                       |
| <b>Dependencies</b>                | Database connection, reporting modules.                       |
| <b>Priority</b>                    | High                                                          |

Table 41 Functional Requirement#9

|                                    |                                                                                    |
|------------------------------------|------------------------------------------------------------------------------------|
| <b>Identifier</b>                  | Functional Requirement#9                                                           |
| <b>Title</b>                       | Record fee payment                                                                 |
| <b>Requirement</b>                 | The system shall allow authorized users to record and manage student fee payments. |
| <b>Source</b>                      | Only the Administrator can record the payment.                                     |
| <b>Rationale</b>                   | To maintain Accurate and current financial records.                                |
| <b>Business Rule (if required)</b> | Only the administrator can record a payment.                                       |
| <b>Dependencies</b>                | Database connection                                                                |



|                 |      |
|-----------------|------|
| <b>Priority</b> | High |
|-----------------|------|

Table 42 Functional Requirement#10

|                                    |                                                                                         |
|------------------------------------|-----------------------------------------------------------------------------------------|
| <b>Identifier</b>                  | Functional Requirement#10                                                               |
| <b>Title</b>                       | View Enrolled Subjects                                                                  |
| <b>Requirement</b>                 | Students can view a list of all subjects they are currently enrolled in.                |
| <b>Source</b>                      | KIPS management                                                                         |
| <b>Rationale</b>                   | To allow students to track their academic progress and access subject-specific content. |
| <b>Business Rule (if required)</b> | Student must be actively enrolled in at least one subject.                              |
| <b>Dependencies</b>                | FR#2(Student Login)                                                                     |
| <b>Priority</b>                    | High                                                                                    |

Table 43 Functional Requirement#11

|                                    |                                                                                                                                        |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Identifier</b>                  | Functional Requirement#11                                                                                                              |
| <b>Title</b>                       | Attend online classes                                                                                                                  |
| <b>Requirement</b>                 | The system shall allow students to view the scheduled online and join the class using the provided meeting link at the scheduled time. |
| <b>Source</b>                      | KIPS requirement, student feedback                                                                                                     |
| <b>Rationale</b>                   | Student need a reliable way to attend online classes.                                                                                  |
| <b>Business Rule (if required)</b> | Student must be registered in the class. Meeting link is accessible only during class time. Stable internet connection is required.    |
| <b>Dependencies</b>                | FR#1                                                                                                                                   |



|                 |      |
|-----------------|------|
| <b>Priority</b> | High |
|-----------------|------|

Table 44 Functional Requirement#12

|                                    |                                                                                       |
|------------------------------------|---------------------------------------------------------------------------------------|
| <b>Identifier</b>                  | Functional Requirement#12                                                             |
| <b>Title</b>                       | View and attempt Quizzes and assignments                                              |
| <b>Requirement</b>                 | Students accesses and attempts interactive quizzes and assignments within the system. |
| <b>Source</b>                      | KIPS requirement, App Features                                                        |
| <b>Rationale</b>                   | To allow students to test their understanding and self-access effectively.            |
| <b>Business Rule (if required)</b> | Student must be registered and login.                                                 |
| <b>Dependencies</b>                | FR#9(view enrolled subjects)                                                          |
| <b>Priority</b>                    | High                                                                                  |

Table 45 Functional Requirement#13

|                                    |                                                                 |
|------------------------------------|-----------------------------------------------------------------|
| <b>Identifier</b>                  | Functional Requirement#13                                       |
| <b>Title</b>                       | Download Study Material                                         |
| <b>Requirement</b>                 | Students logs in and downloads study material in PDF format.    |
| <b>Source</b>                      | KIPS requirement, Student Feedback                              |
| <b>Rationale</b>                   | To provide students with offline access to educational content. |
| <b>Business Rule (if required)</b> | Student must be registered and login.                           |





|                     |                          |
|---------------------|--------------------------|
| <b>Dependencies</b> | FR#5(View Quizzes/Tests) |
| <b>Priority</b>     | High                     |

Table 46 Functional Requirement#14

|                                    |                                                                           |
|------------------------------------|---------------------------------------------------------------------------|
| <b>Identifier</b>                  | Functional Requirement#14                                                 |
| <b>Title</b>                       | Announcements                                                             |
| <b>Requirement</b>                 | System sends announcements to students about classes, quizzes and result. |
| <b>Source</b>                      | KIPS requirement, Student Feedback                                        |
| <b>Rationale</b>                   | To keep students informed and engaged with their academic progress.       |
| <b>Business Rule (if required)</b> | Student must be enrolled in a course.                                     |
| <b>Dependencies</b>                | FR#7(View Fee Status)                                                     |
| <b>Priority</b>                    | High                                                                      |

Table 47 Functional Requirement#15

|                                    |                                                                                                   |
|------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>Identifier</b>                  | Functional Requirement#15                                                                         |
| <b>Title</b>                       | Classes Timetable                                                                                 |
| <b>Requirement</b>                 | Display a weekly timetable for each class. Showing the subjects, time, date and assigned teacher. |
| <b>Source</b>                      | Kips requirement                                                                                  |
| <b>Rationale</b>                   | Students and teachers need a clear schedule to plan their activities.                             |
| <b>Business Rule (if required)</b> | Classes can't be scheduled during lunch breaks.                                                   |
| <b>Dependencies</b>                | FR#1 (Teacher Login)                                                                              |



|                 |      |
|-----------------|------|
| <b>Priority</b> | High |
|-----------------|------|

Table 48 Functional Requirement#16

|                                    |                                                                         |
|------------------------------------|-------------------------------------------------------------------------|
| <b>Identifier</b>                  | Functional Requirement#16                                               |
| <b>Title</b>                       | Upload Study Material                                                   |
| <b>Requirement</b>                 | Upload study material in pdf form to a specific course or class.        |
| <b>Source</b>                      | KIPS requirement, teachers feedback                                     |
| <b>Rationale</b>                   | Teachers need a mechanism to share resources with students efficiently. |
| <b>Business Rule (if required)</b> | Only teachers can access the upload functionality.                      |
| <b>Dependencies</b>                | FR#2 (Class Timetable)                                                  |
| <b>Priority</b>                    | High                                                                    |

Table 49 Functional Requirement#17

|                                    |                                                                                      |
|------------------------------------|--------------------------------------------------------------------------------------|
| <b>Identifier</b>                  | Functional Requirement#17                                                            |
| <b>Title</b>                       | Create Quizzes                                                                       |
| <b>Requirement</b>                 | Create new quizzes, add questions, set time limits and upload them to specific class |
| <b>Source</b>                      | KIPS requirement                                                                     |
| <b>Rationale</b>                   | Teachers need a tool to assess student knowledge efficiently.                        |
| <b>Business Rule (if required)</b> | At least 2 questions must and maximum time 30 minutes and maximum file size is 50MB. |
| <b>Dependencies</b>                | FR#3 (Upload Study Material)                                                         |
| <b>Priority</b>                    | High                                                                                 |



Table 50 Functional Requirement#18

|                                    |                                                                                                                                 |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Identifier</b>                  | Functional Requirement#18                                                                                                       |
| <b>Title</b>                       | Create link for online class                                                                                                    |
| <b>Requirement</b>                 | System shall allow the teacher to create an online class by entering class detail and adding a link and share it with students. |
| <b>Source</b>                      | KIPS requirement                                                                                                                |
| <b>Rationale</b>                   | Teacher need an easy way to schedule and conduct online classes.                                                                |
| <b>Business Rule (if required)</b> | Meeting link must be valid. Class date and time cannot be in the past. Only assigned can create links for their classes.        |
| <b>Dependencies</b>                | FR#1                                                                                                                            |
| <b>Priority</b>                    | High                                                                                                                            |

Table 51 Functional Requirement#19

|                                    |                                                                             |
|------------------------------------|-----------------------------------------------------------------------------|
| <b>Identifier</b>                  | Functional Requirement#19                                                   |
| <b>Title</b>                       | Manage Student Performance                                                  |
| <b>Requirement</b>                 | Record student performance data (grades, comments) for specific assignment. |
| <b>Source</b>                      | KIPS requirement, teacher feedback.                                         |
| <b>Rationale</b>                   | To track student progress and provide feedback.                             |
| <b>Business Rule (if required)</b> | Data entry must be validated against defined grade scales.                  |
| <b>Dependencies</b>                | FR#4 (Create Online Quizzes)                                                |
| <b>Priority</b>                    | High                                                                        |



Table 52 Functional Requirement#20

|                                    |                                                                                                   |
|------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>Identifier</b>                  | Functional Requirement#20                                                                         |
| <b>Title</b>                       | Communication with Parents                                                                        |
| <b>Requirement</b>                 | Send text messages or emails to parents of students.                                              |
| <b>Source</b>                      | KIPS requirement, parent feedback                                                                 |
| <b>Rationale</b>                   | Timely communication between teachers and parents regarding progress, behavior and announcements. |
| <b>Business Rule (if required)</b> | Messages must be logged.                                                                          |
| <b>Dependencies</b>                | FR#5 (Manage Student Performance)                                                                 |
| <b>Priority</b>                    | High                                                                                              |

Table 53 Functional Requirement#21

|                                    |                                                                                    |
|------------------------------------|------------------------------------------------------------------------------------|
| <b>Identifier</b>                  | Functional Requirement#21                                                          |
| <b>Title</b>                       | Upload and view Assignments                                                        |
| <b>Requirement</b>                 | Upload an assignment to students and view previous assignments.                    |
| <b>Source</b>                      | KIPS requirement, teacher feedback                                                 |
| <b>Rationale</b>                   | Mechanism to distribute assignments to students and manage them within the system. |
| <b>Business Rule (if required)</b> | Maximum file size is 50MB and only PDF file format.                                |
| <b>Dependencies</b>                | FR#6 (Communication with Parents)                                                  |
| <b>Priority</b>                    | High                                                                               |

Table 54 Functional Requirement#22

|                   |                           |
|-------------------|---------------------------|
| <b>Identifier</b> | Functional Requirement#22 |
|-------------------|---------------------------|



|                                    |                                                                            |
|------------------------------------|----------------------------------------------------------------------------|
| <b>Title</b>                       | Link parent to child                                                       |
| <b>Requirement</b>                 | System allows parents to securely link their account to a child's account. |
| <b>Source</b>                      | KIPS requirement, parent Feedback                                          |
| <b>Rationale</b>                   | To ensure proper account management and parental oversight.                |
| <b>Business Rule (if required)</b> | Parent and child must both have valid accounts                             |
| <b>Dependencies</b>                | Parent and child account registration requirements                         |
| <b>Priority</b>                    | High                                                                       |

Table 55 Functional Requirement#23

|                                    |                                                                                              |
|------------------------------------|----------------------------------------------------------------------------------------------|
| <b>Identifier</b>                  | Functional Requirement#23                                                                    |
| <b>Title</b>                       | View child profile                                                                           |
| <b>Requirement</b>                 | System allows parents to securely view their child's profile information                     |
| <b>Source</b>                      | KIPS requirement, parent Feedback, system design                                             |
| <b>Rationale</b>                   | Allow parents to monitor and manage their child's information and settings within the system |
| <b>Business Rule (if required)</b> | Parent must be logged in and linked to the child's profile.                                  |
| <b>Dependencies</b>                | Parent login, child profile creation, link accounts                                          |
| <b>Priority</b>                    | High                                                                                         |

Table 56 Functional Requirement#24

|                   |                           |
|-------------------|---------------------------|
| <b>Identifier</b> | Functional Requirement#24 |
|-------------------|---------------------------|



|                                    |                                                                            |
|------------------------------------|----------------------------------------------------------------------------|
| <b>Title</b>                       | View results                                                               |
| <b>Requirement</b>                 | System allows parents and students to securely view their academic results |
| <b>Source</b>                      | KIPS requirement, parent Feedback                                          |
| <b>Rationale</b>                   | To provide parents with timely updates on their child's academic progress. |
| <b>Business Rule (if required)</b> | Parent must be logged in and have a child associated with their account.   |
| <b>Dependencies</b>                | Payroll login requirement, student result data availability                |
| <b>Priority</b>                    | High                                                                       |

Table 57 Functional Requirement#25

|                                    |                                                                                                                  |
|------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Identifier</b>                  | Functional Requirement#25                                                                                        |
| <b>Title</b>                       | Notification for Fee payment                                                                                     |
| <b>Requirement</b>                 | System allows parent to securely pay fees for their children                                                     |
| <b>Source</b>                      | KIPS requirement, parent Feedback                                                                                |
| <b>Rationale</b>                   | To provide a convenient and secure method for parents to settle financial obligations and ensure timely payments |
| <b>Business Rule (if required)</b> | Parent must have a valid username and password and a valid payment method                                        |
| <b>Dependencies</b>                | Parent login requirement, payment gateway integration                                                            |
| <b>Priority</b>                    | High                                                                                                             |

Table 58 Functional Requirement#26

|                    |                                                                                          |
|--------------------|------------------------------------------------------------------------------------------|
| <b>Identifier</b>  | Functional Requirement#26                                                                |
| <b>Title</b>       | View fee history and status                                                              |
| <b>Requirement</b> | System allows parents to securely view their child's payment history and current status. |
| <b>Source</b>      | KIPS requirement, parent Feedback, Student feedback                                      |



|                                    |                                                                                 |
|------------------------------------|---------------------------------------------------------------------------------|
| <b>Rationale</b>                   | To ensure transparency in financial matters and allow parents to track payments |
| <b>Business Rule (if required)</b> | Parent must be logged in and linked to a student account                        |
| <b>Dependencies</b>                | Parent login, student account, fee payment system                               |
| <b>Priority</b>                    | High                                                                            |

Table 59 Functional Requirement #27

|                      |                                                                               |
|----------------------|-------------------------------------------------------------------------------|
| <b>Identifier</b>    | Functional Requirement #27                                                    |
| <b>Title</b>         | Give Feedback and Review                                                      |
| <b>Requirement</b>   | Student and parent shall provide feedback and reviews about their experience. |
| <b>Source</b>        | Student, parent                                                               |
| <b>Rationale</b>     | Improve System quality                                                        |
| <b>Business Rule</b> | Inappropriate review must be removed                                          |
| <b>Dependencies</b>  | None                                                                          |
| <b>Priorities</b>    | Medium                                                                        |

Table 60 Functional Requirement#28

|                                    |                                                                 |
|------------------------------------|-----------------------------------------------------------------|
| <b>Identifier</b>                  | Functional Requirement#28                                       |
| <b>Title</b>                       | Log out                                                         |
| <b>Requirement</b>                 | System allows users to securely log out from this session       |
| <b>Source</b>                      | KIPS requirement, Users Feedback                                |
| <b>Rationale</b>                   | To ensure account security and privacy and manage system access |
| <b>Business Rule (if required)</b> | None                                                            |
| <b>Dependencies</b>                | User login requirements                                         |
| <b>Priority</b>                    | High                                                            |

### 3.4. Non-functional requirements

The non-functional requirements of **KIPS COACHING KHARIAN** are followed below:

#### 3.4.1. Reliability requirements



The system must ensure minimum downtime and available to the users during operational hours. Users need easy access to the platform for all coaching activities. A reliable system ensures continuity of service during all scheduled operational hours.

#### **3.4.2. Usability requirements**

The interface must be simple, easy to navigate and used on mobile devices. Clear error message will guide users that is human readable message (e.g. username you entered is not recognized. Please check your spelling).

#### **3.4.3. Performance requirements**

The system will load quickly and response in 4 or 5 second if the user has good internet connection. It supports the users to perform their tasks.

#### **3.4.4. Security requirements**

All the user's data including personal information, academic records and personal information must be stored securely. The system should have clear access controls defining which action each type of user can perform.

### **3.5. External interface requirements**

The external interface requirements of our system are given below:

#### **3.5.1. User interface requirements**

The application is developed for the android mobile platform using Android Studio and Java. The UI is designed to be responsive across various android device screen size and resolutions. User interactions like data submission to Firebase. Error message is displayed for invalid inputs.

#### **3.5.2. System interface requirements**

The system integrates with Firebase authentication to manage user login, registrations and session management. The system will facilitate joining classes through a teacher shared link.

#### **3.5.3. Hardware interface requirements**

The system run on Android smartphones. The system may require access to standard mobile hardware components such as touch screen for user input and network connectivity for database connectivity.





### 3.6. Summary

Requirement Analysis provided an insightful overview of project. We analyzed the requirements that are required by our project by using several requirement identifying techniques i.e. use case diagram, story board etc. We also provided a detailed overview of our use cases, which clearly describe each use case along with its activities. After describing the use cases, we also explained the FRs of our system that must be fulfilled to complete system's functionality. By describing these requirements in a structured format, we are confident that no ambiguity is left in the requirement analysis of our project and of course, we cannot forget the NFRs of any project, so we have described them as well, along with the external interface requirements. The project is an Android application developed using java within Android Studio. The Firebase is used as the backend as a service for its real time database and authentication. The non-functional requirements focus on reliability, usability and security. The external interface is the android user interface, the Firebase APIs for data management and standard mobile hardware component.



## **Chapter 4**

### **Design and Architecture**



## 4. System Design

### 4.1 Design Considerations

#### Assumptions

Students, teachers, admins and parents will have a stable internet connection to access the app and manage information. It will provide accurate and up-to-date course schedules, student records and announcements. The application will be used primarily on Android devices and will expand in the future for iOS devices. The payment gateways for fee submission will be functional and available all the time.

#### Dependencies

Android Studio and other libraries for the development of apps. Third-party payment gateways will be used for transactions (fee submission). The app's backend system uses a database to keep all its information organized (student grades, teacher schedules, etc.). Google Maps to provide location information for the physical campus/branches.

#### Limitations

The primary limitation is that Kips Coaching Kharian is a Kharian-based service and physical classes are conducted only in Kharian City. The app requires an active internet connection to work properly. Payment failures may arise due to issues with third-party payment gateways.

#### Risks

There is a risk, if the project does not satisfy the requirements of the management team at Kips Coaching Kharian. Users (students, parents, teachers, admin) can cause errors by entering incorrect data or selecting the wrong options. If a user cancels a subject or requests a refund at a time that is against the terms and conditions, it can cause conflicts. The project might face resistance from traditional administrative staff who do not want to upgrade to the online system.

### 4.2 Requirement Traceability Matrix:

#### Traceability Matrix:

Table 61 Traceability matrix

| Requirement ID | Requirement Description | Design specification     | Code                    |
|----------------|-------------------------|--------------------------|-------------------------|
| R1             | Admin Login             | Login component          | AdminLoginActivity.java |
| R2             | Reset Password          | Reset password component | ResetPassword.java      |



|     |                                          |                                        |                                    |
|-----|------------------------------------------|----------------------------------------|------------------------------------|
| R3  | Add, update and delete users             | Manage user                            | ManageUserActivity.java            |
| R4  | Approve student and parent registration  | Manage student and parent registration | ApproveRegistrationActivity.java   |
| R5  | Manage announcements                     | Announcement manager                   | AnnouncementManagerActivity.java   |
| R6  | View system report                       | System report viewer                   | SystemReportActivity.java          |
| R7  | Record fee payment                       | Fee payment manager                    | FeePaymentActivity.java            |
| R8  | Student Register                         | Register component                     | StudentRegisterActivity.java       |
| R9  | Student Login                            | Login component                        | StudentLoginActivity.java          |
| R10 | View enrolled subjects                   | Enrolled subjects viewer               | EnrolledSubjectsActivity.java      |
| R11 | Download study material                  | Study material downloader              | StudyMaterialDownloader.java       |
| R12 | View and attempt quizzes and assignments | Quizzes and assignment viewer          | QuizAssignmentViewerActivity.java  |
| R13 | View results and fee history status      | Result and fee history viewer          | FeeHistoryActivity.java            |
| R14 | Receive Notification for fee payment     | Notification server                    | FeeNotificationsActivity.java      |
| R15 | Receive announcements                    | Notification server                    | NotificationActivity.java          |
| R16 | Attempt quizzes                          | Quizzes component                      | QuizActivity.java                  |
| R17 | Teacher login                            | Login component                        | TeacherLoginActivity.java          |
| R18 | Upload study material                    | Study material uploader                | StudyMaterialUploaderActivity.java |
| R19 | Create quizzes                           | Quizzes handler                        | QuizActivity.java                  |
| R20 | Create and share link for online classes | Online classes manager                 | OnlineClassActivity.java           |

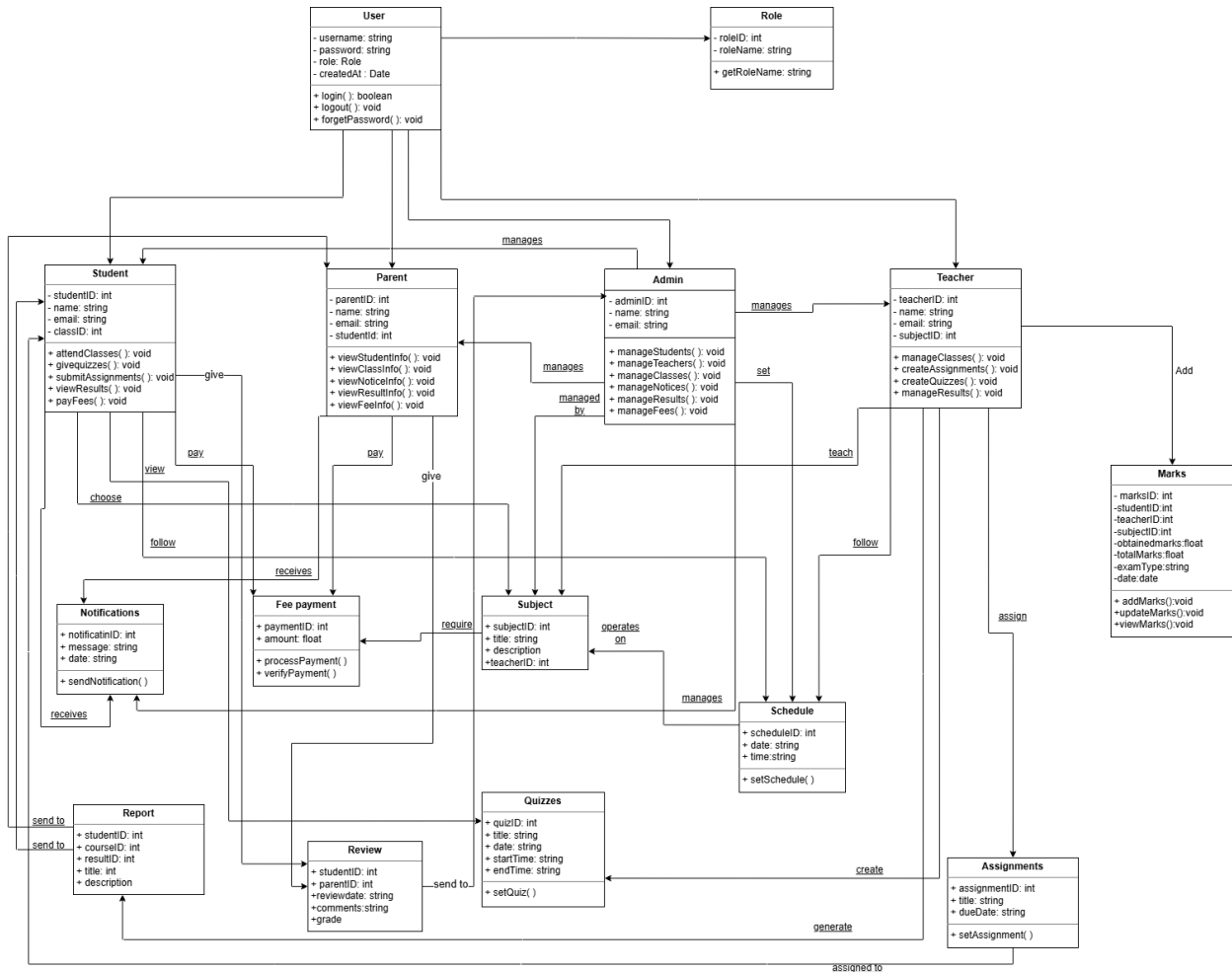


|     |                                     |                                   |                                       |
|-----|-------------------------------------|-----------------------------------|---------------------------------------|
| R21 | Manage student performance          | Student performance manager       | ManageStudentActivity.java            |
| R22 | View and upload assignment          | Assignment viewer                 | AssignmentActivity.java               |
| R23 | Receive announcement/ notifications | Notification server               | AnnouncementActivity.java             |
| R24 | Communication with parents          | Communication with parent manager | CommunicationwithparentsActivity.java |
| R25 | Parent registration                 | Registration component            | ParentregisterActivity.java           |
| R26 | Parent login                        | Login component                   | ParentloginActivity.java              |
| R27 | Link to child account               | Child account manager             | LinktochildActivity.java              |
| R28 | View child account                  | Child account viewer              | Childaccountviewer.java               |
| R29 | View child result                   | Result viewer                     | Resultviewer.java                     |
| R30 | Fee payment and history             | Fee History manager               | Feehistoryviewer.java                 |
| R31 | Fee payment notification            | Notification server               | NotificationActivity.java             |
| R32 | Give review and feedback            | Feedback module                   | FeedbackActivity.java                 |
| R33 | View announcements                  | Announcement manager              | Announcementviewer.java               |
| R34 | Logout                              | Logout manager                    | LogoutActivity.java                   |

### 4.3 Design Models

#### 1.Design Class Diagram (DCD)

The class diagram represents the core structure of the system, defining essential classes such as Students, Teachers, Parents, Classes, Admin along with their attributes and methods. These classes interact with each other to handle class schedule, fee payment, reports etc.





## Admin Sequence diagram

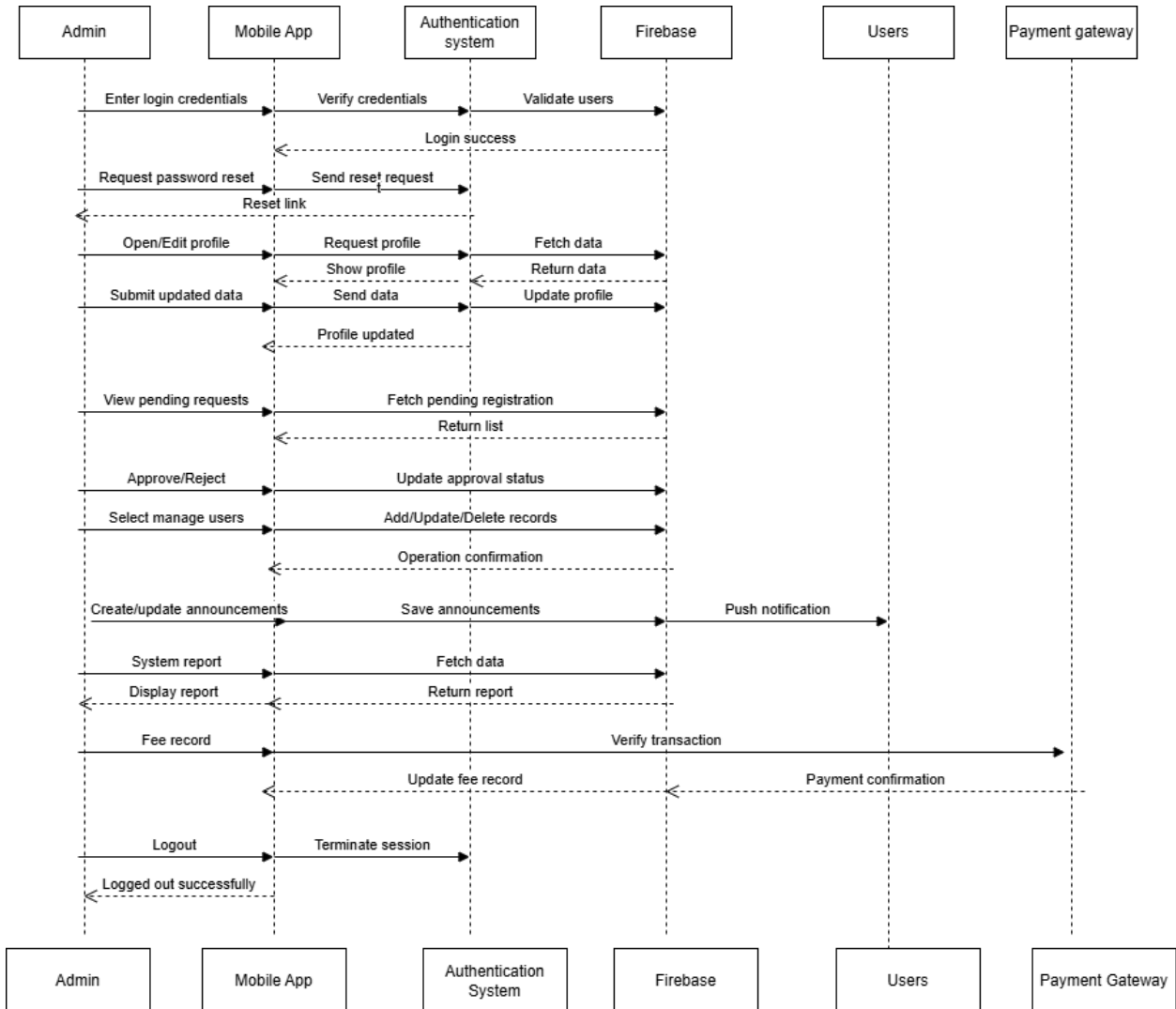


Figure 5: Sequence Diagram (Admin)



## Student Sequence diagram

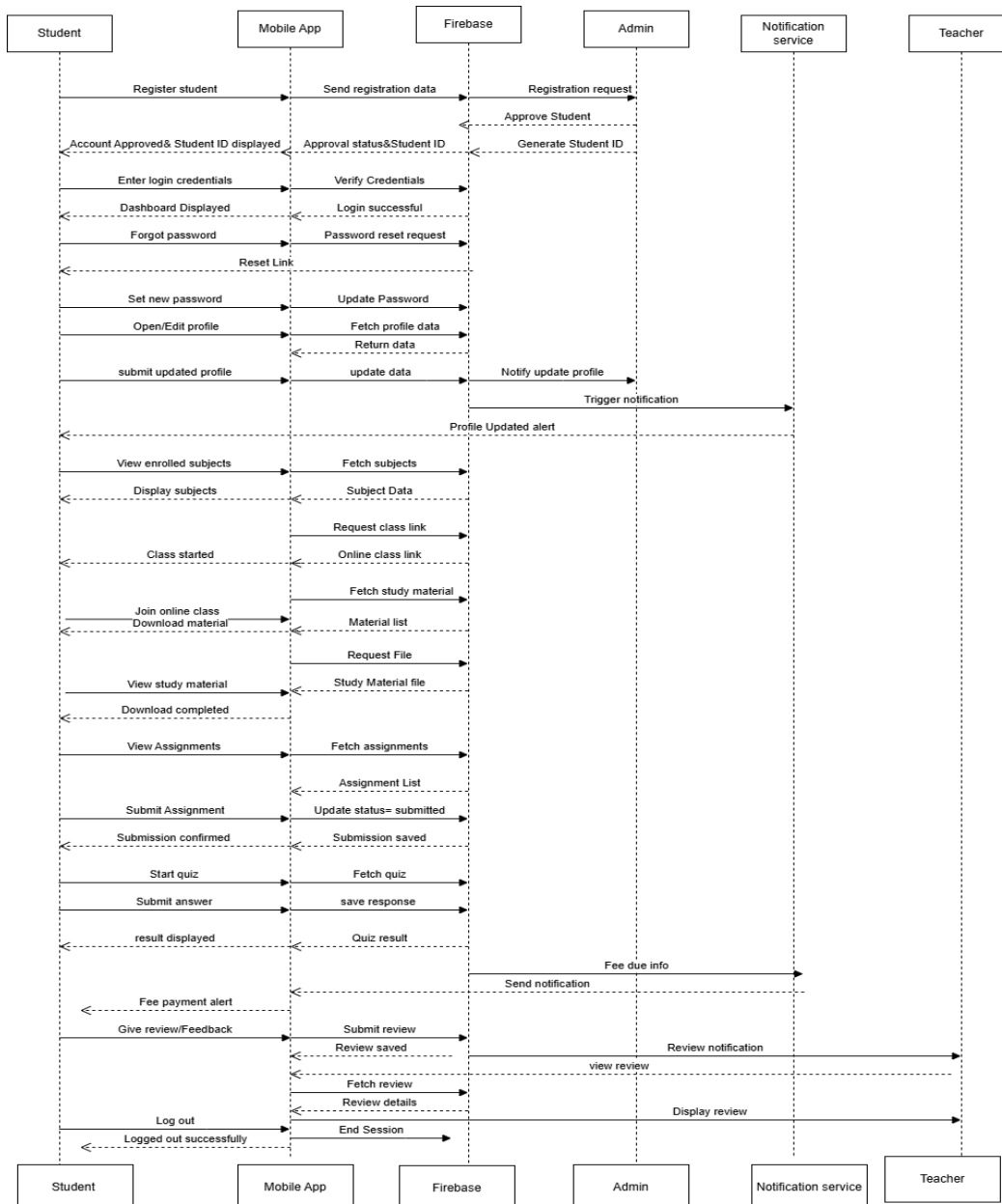


Figure 6: Sequence Diagram (Student)





## Teacher Sequence diagram

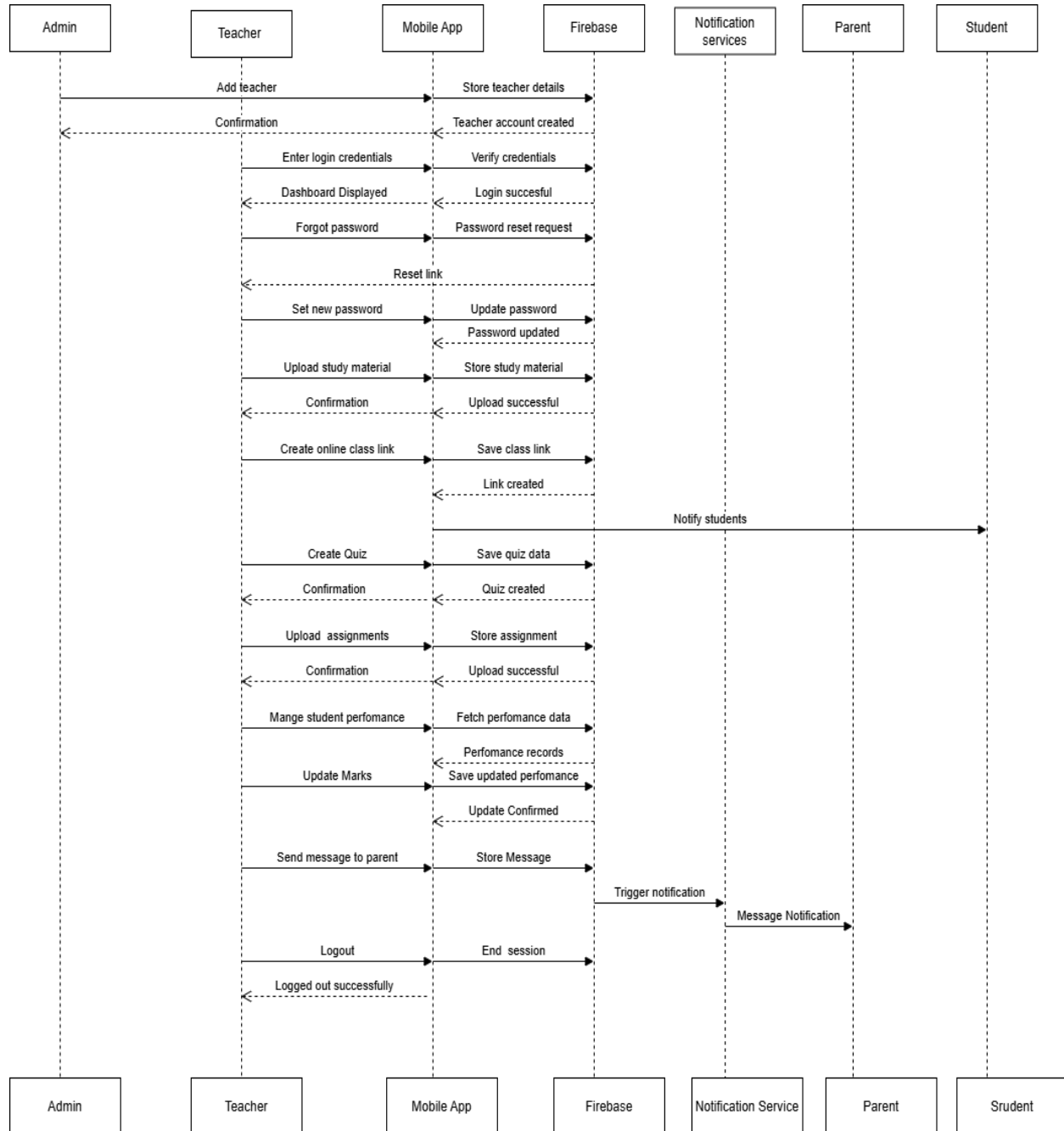


Figure 7: Sequence Diagram (Teacher



## Parent Sequence Diagram



Figure 8: Sequence Diagram (Parents)



### 3.State Transition Diagram

This diagram shows the different states of the KIPS coaching app and the transitions between them. It covers all the student's login from initially opening the app and to register an account, fees payment, timetable checking, view assignments, quizzes and successfully logout from the app.

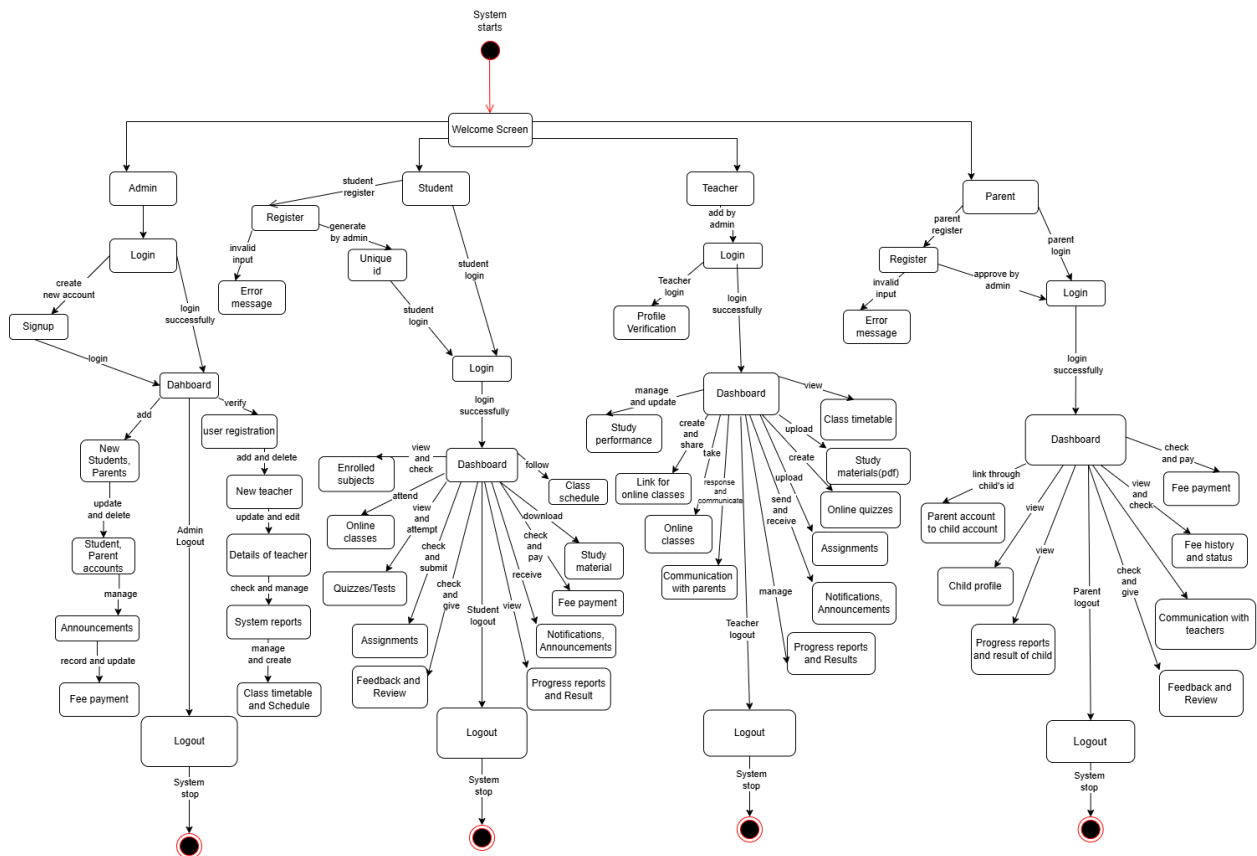


Figure 9: State Diagram

### 4.4 Architectural Diagram

The KIPS Coaching App is built using a 3-tier structure that includes the User Interface (Frontend), Business Logic (Backend) and Database Layer. The frontend is the mobile app that students interact with.

It allows students to login, register, view schedules, access study materials, view assignments, view quizzes, check results, check fee status and notifications. Enables teachers to manage class schedules, access study materials, upload assignments and quizzes, student progress and send announcements. Provide admins with tools for user management (students, teachers, parents),



course management, fee tracking and announcements. Allows parents to monitor their child progress, grades, fees and communication with teachers.

When a user does something in the app, the backend takes care of the main work, like checking login details, fees handling, assignments and quizzes. This backend then talks to the database, which is built using firebase. The database saves important data. The backend also connects other services like fee payment (such as payfast) to send messages and complete payments safely. This keeps everything clean and organized.

## 1.Tier Architecture Diagram

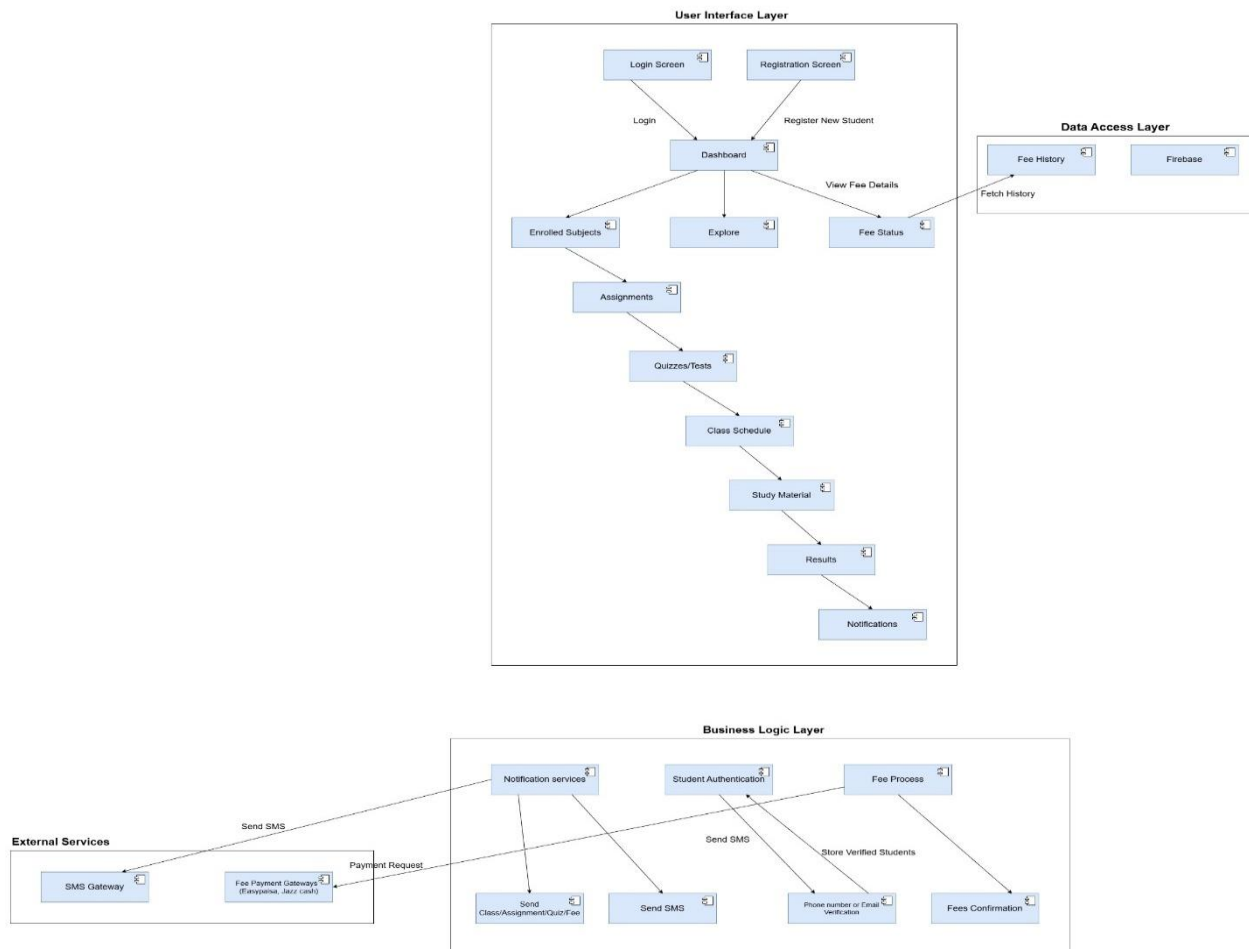
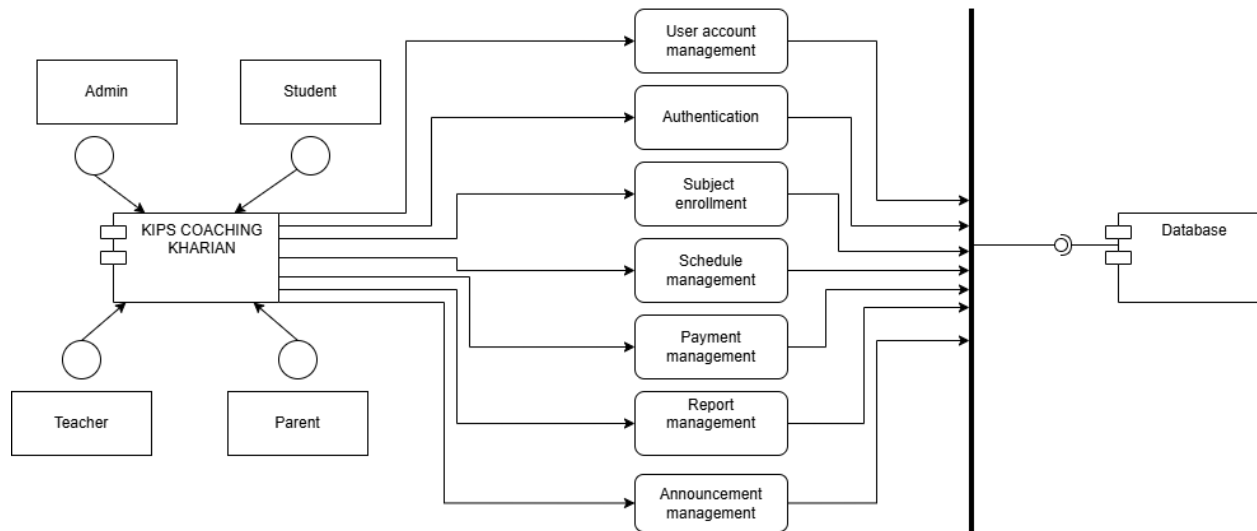


Figure 10: Architecture Diagram



## 2.UML component diagram:



*Figure 11: UML component diagram*

## 4.5 Data design

Data design shows how the information in the KIPS COACHING KHARIAN is stored, organized and managed. It uses firebase to store and organize data.



## Erd Schema:

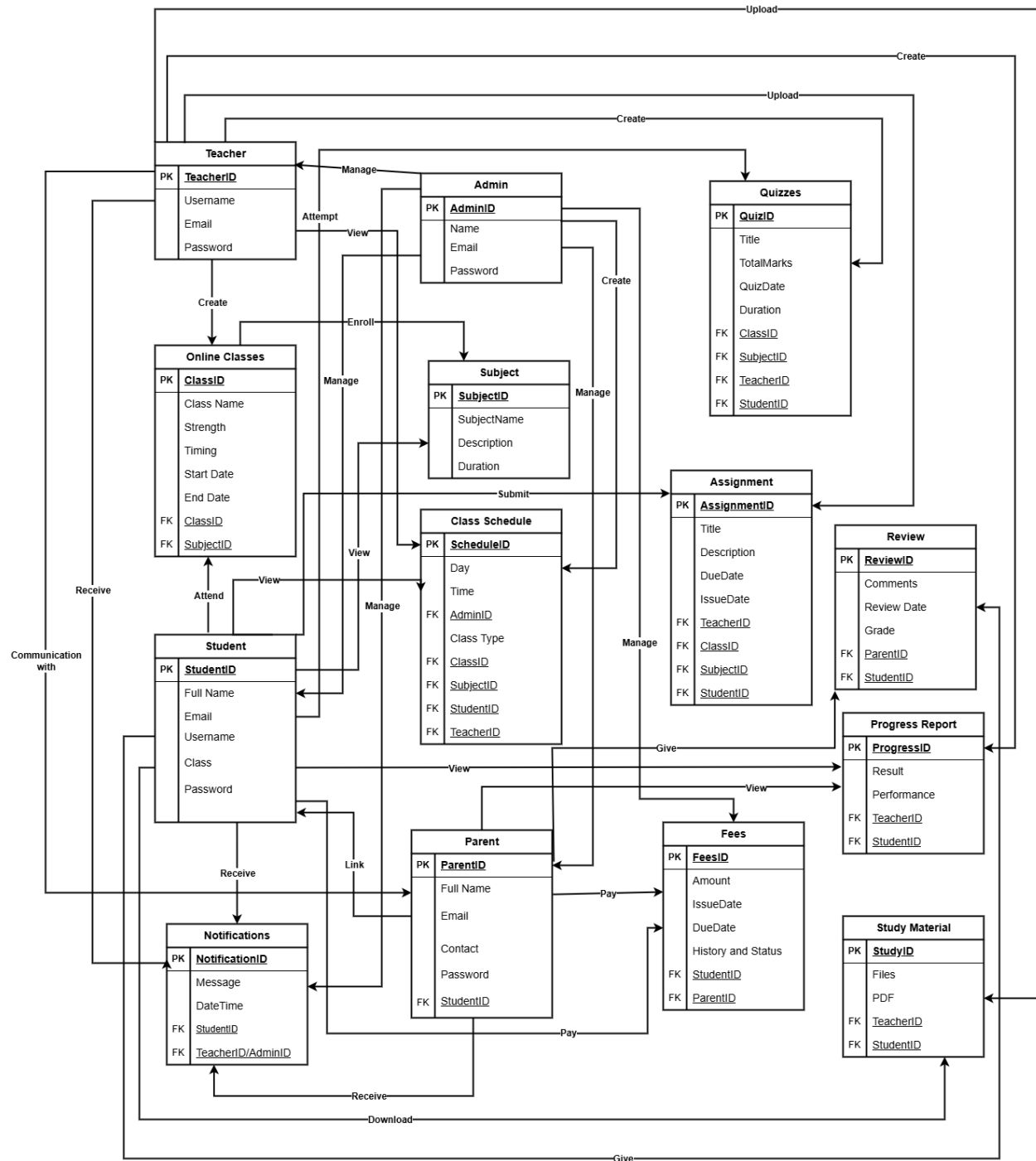


Figure 12: ERD Schema



## 2.Data dictionary

This table contains the alphabetical list of system entities with their type and descriptions for **KIPS COACHING KHARIAN APP**.

Table 62: Data Dictionary

| Terminology              | Data type      | Description                          |
|--------------------------|----------------|--------------------------------------|
| Amount                   | Decimal (10,2) | Fee amount                           |
| Announcement_description | Text           | Announcement details                 |
| Announcement_id          | Int            | Announcement unique ID               |
| Announcement_title       | Varchar (150)  | Announcement title                   |
| Assignment_id            | Int            | Assignment unique ID                 |
| Assignment_marks         | Int            | Assignment obtained marks            |
| Assignment_status        | Varchar (20)   | Pending / Submitted                  |
| Class                    | Varchar (50)   | Student class                        |
| Created_at               | Datetime       | Account creation date                |
| Due_date                 | Date           | Assignment deadline                  |
| Email                    | Varchar (150)  | User login email                     |
| Fee_id                   | Int            | Fee record ID                        |
| File_url                 | Varchar (255)  | Study material or assignment file    |
| Last_login               | Datetime       | Last login timestamp                 |
| Login_id                 | Int            | Login record ID                      |
| Material_id              | Int            | Study material ID                    |
| Message_id               | Int            | Communication message ID             |
| Name                     | Varchar (100)  | Name of Admin/Teacher/Student/Parent |
| Notification_id          | Int            | Notification ID                      |
| Notification_message     | Text           | Notification content                 |
| Password                 | Varchar (255)  | Encrypted password                   |
| Payment_date             | Date           | Fee payment date                     |
| Payment_method           | Varchar (50)   | Cash / Online                        |
| Phone                    | Varchar (15)   | Contact number                       |
| Quiz_id                  | Int            | Quiz ID                              |
| Quiz_marks               | Int            | Obtained quiz marks                  |



|             |               |                                         |
|-------------|---------------|-----------------------------------------|
| Remarks     | Varchar (255) | Teacher remarks                         |
| Register_id | Int           | Registration ID                         |
| Role        | Varchar (50)  | Admin / Teacher / Student / Parent role |
| Status      | Varchar (20)  | Paid / Unpaid                           |
| Student_id  | Int           | Student unique ID                       |
| Subject     | Varchar (100) | Teaching subject                        |
| Teacher_id  | Int           | Teacher ID                              |
| Title       | Varchar (150) | Quiz / Assignment / Material title      |
| Total_marks | Int           | Quiz total marks                        |
| Upload_date | Datetime      | Material upload date                    |
| User_id     | Int           | General user ID                         |
| User_name   | Varchar (100) | Registered user name                    |
| User_type   | Varchar (20)  | Admin / Teacher / Student / Parent      |

## 4.6 User interfaces design

### 4.6.1 Screen Images

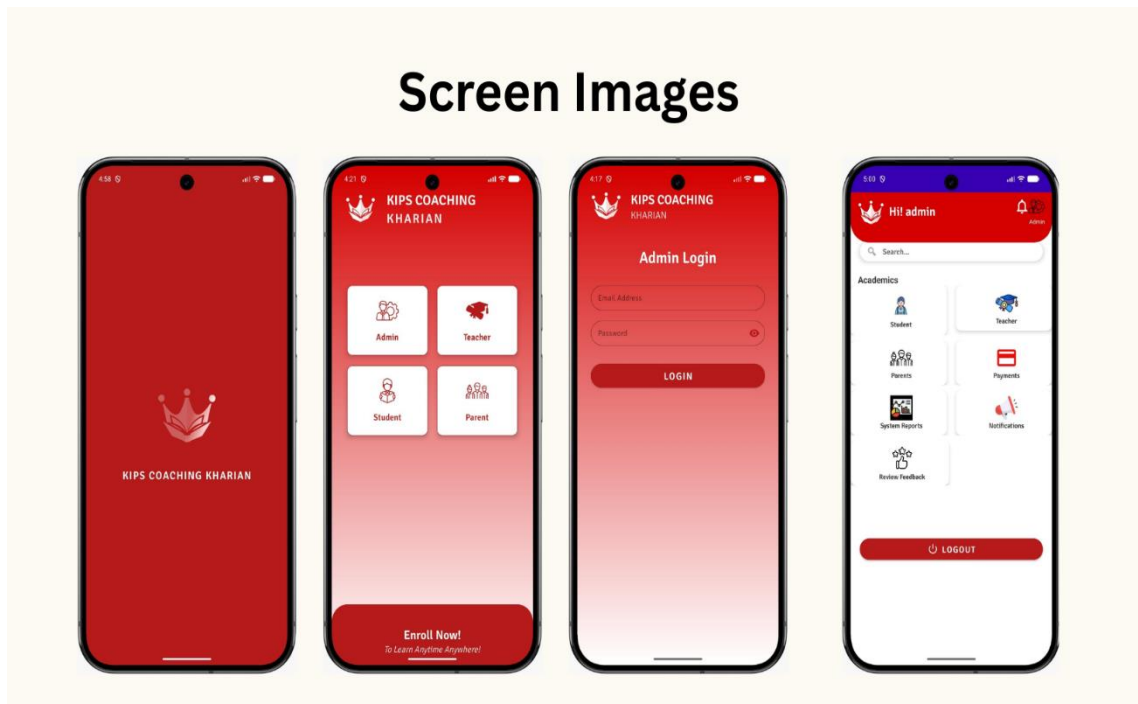


Figure 13: Screen image





## Screen Images

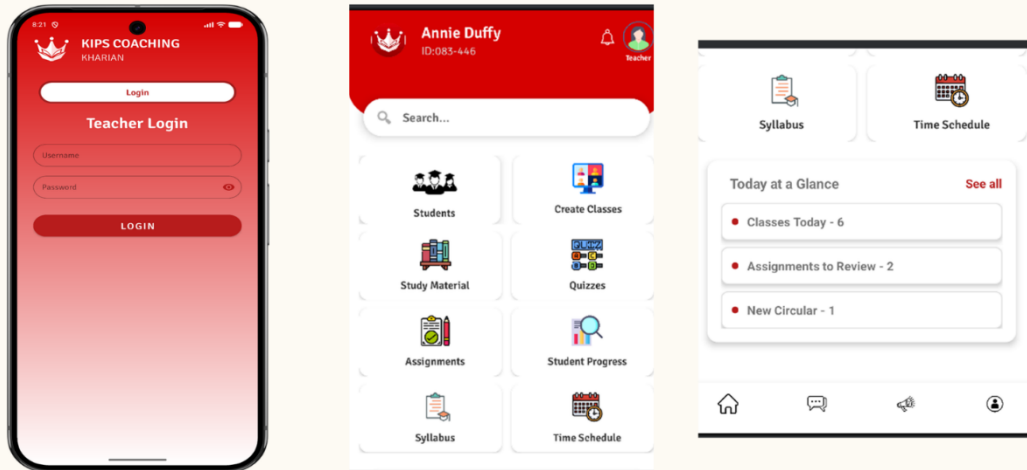


Figure 14: Screen image

## Screen Images

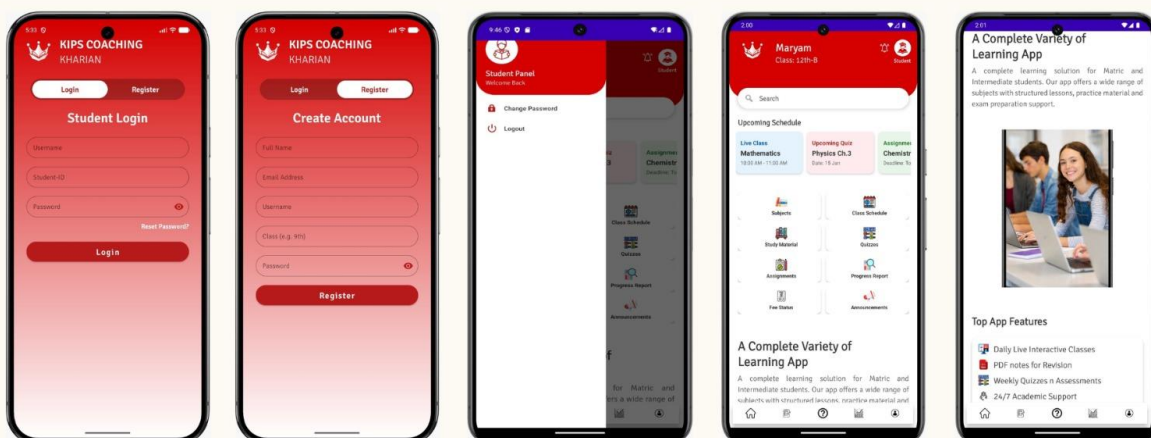


Figure 15: Screen image



## Screen Images

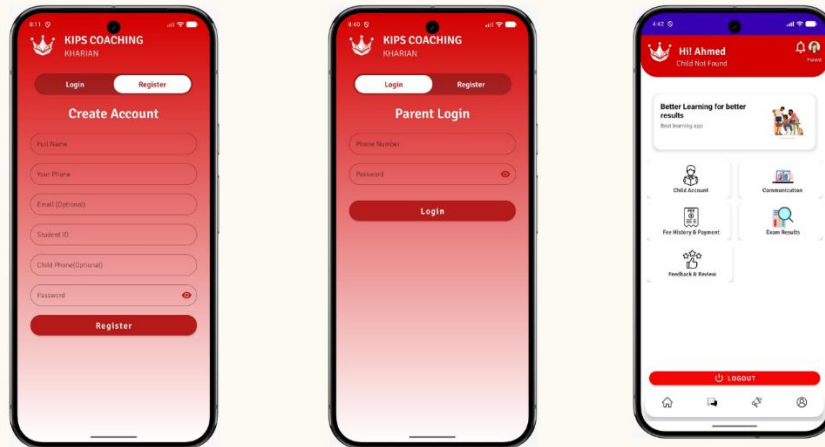


Figure 16: Screen image

### 4.6.2 Screen Objects and Actions:

This section details the screen components and the functions they perform for use cases:

Table 63: Screen object (Live classes)

| Screen Object        | Action                                                                       |
|----------------------|------------------------------------------------------------------------------|
| Teacher dashboard    | Create account and access teacher dashboard then manage live classes section |
| Create live class    | Teacher creates live class                                                   |
| Start/end live class | Teacher starts the live class and end it after completion                    |
| Student dashboard    | Student logs in and view scheduled live class                                |
| Attend live class    | Student attend live class and exits when the class end                       |



Table 64: Screen object (Notifications and announcements)

| Screen Object                    | Action                                            |
|----------------------------------|---------------------------------------------------|
| Admin dashboard                  | Access and manage notifications and announcement  |
| Create notification/announcement | Admin create notifications and announcements      |
| Send notifications               | Send notifications to student, teacher and parent |
| View/receive notifications       | View/receive notifications and announcements      |

Table 65: Screen object (Fee payment and confirmation)

| Screen Object              | Action                                                            |
|----------------------------|-------------------------------------------------------------------|
| Student/parent dashboard   | Access the fee payment section                                    |
| Fee details                | Students and Parents view fee amount, due date and payment status |
| Enter payment details      | Provide necessary information                                     |
| Pay button                 | Complete the fee payment transaction                              |
| Payment confirmation alert | In app confirmation payment notification sent                     |

## 4.7 Behavioral Model

### 1.Sequence Diagram

The sequence diagram helped visualize how different parts of the system, like the app, database and payment gateways are work together for booking of guider and hotels.



## Admin Sequence diagram

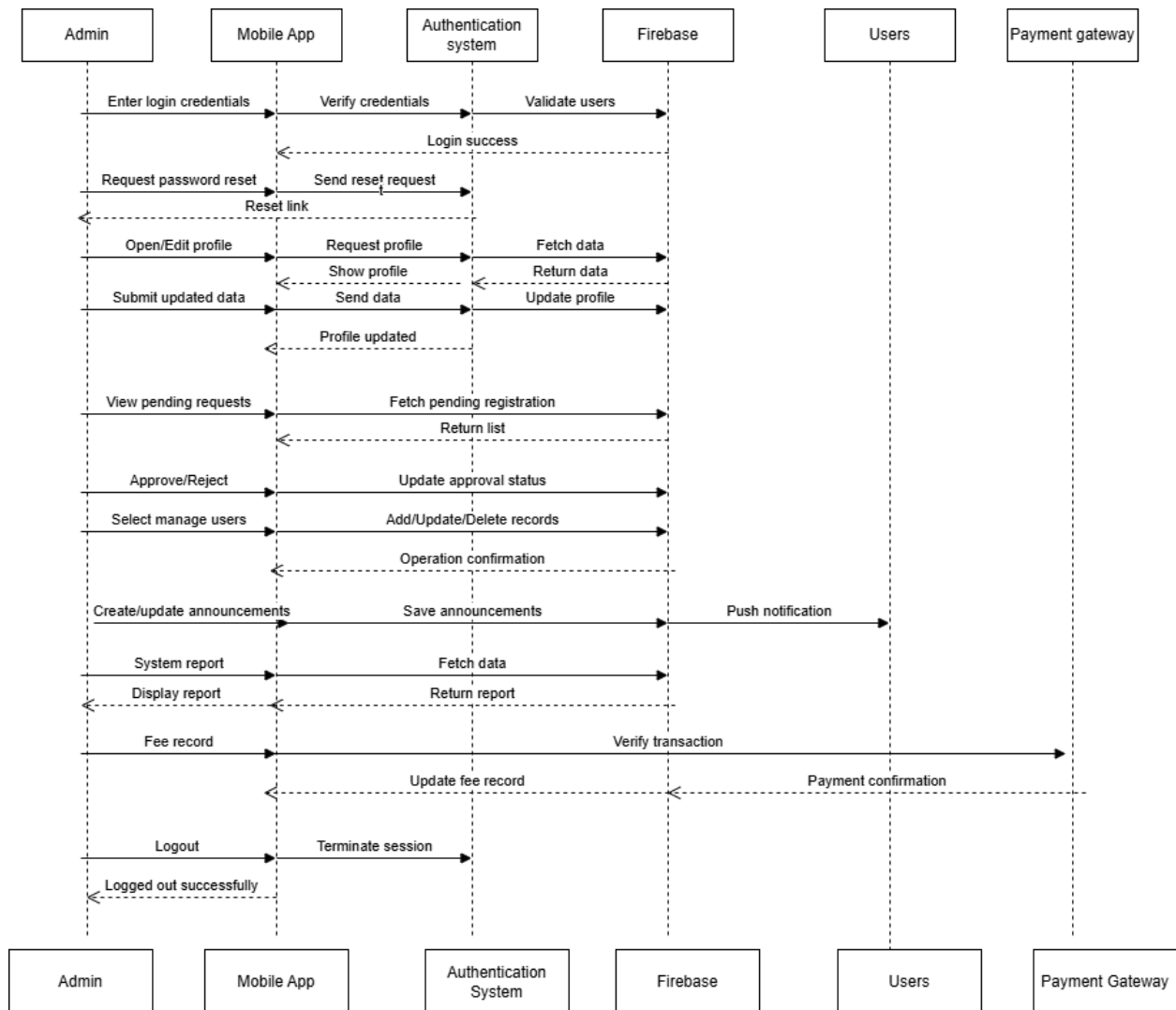


Figure 17: Sequence Diagram (Admin)



## Student Sequence diagram

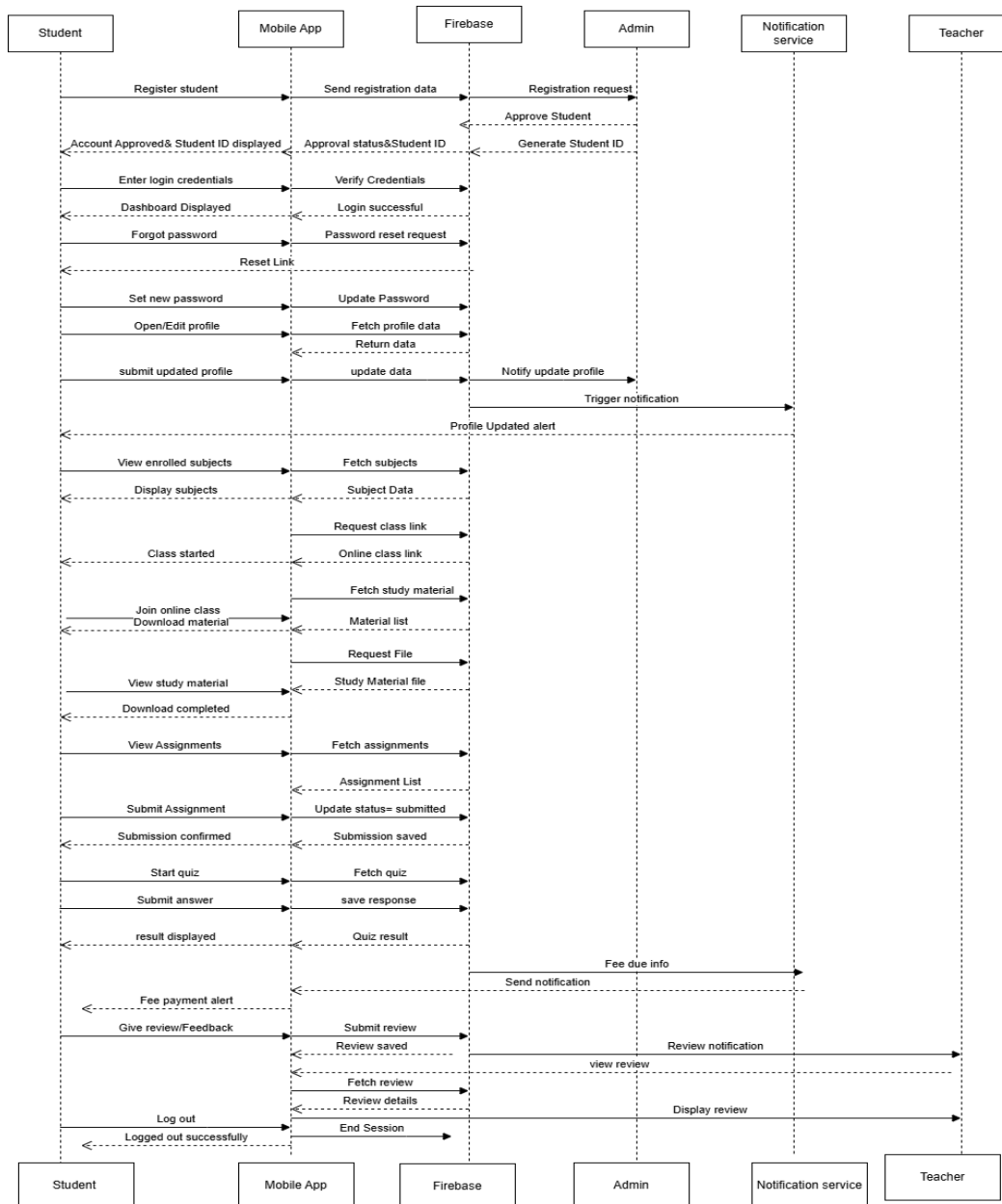


Figure 18: Sequence Diagram (Student)



## Teacher Sequence diagram

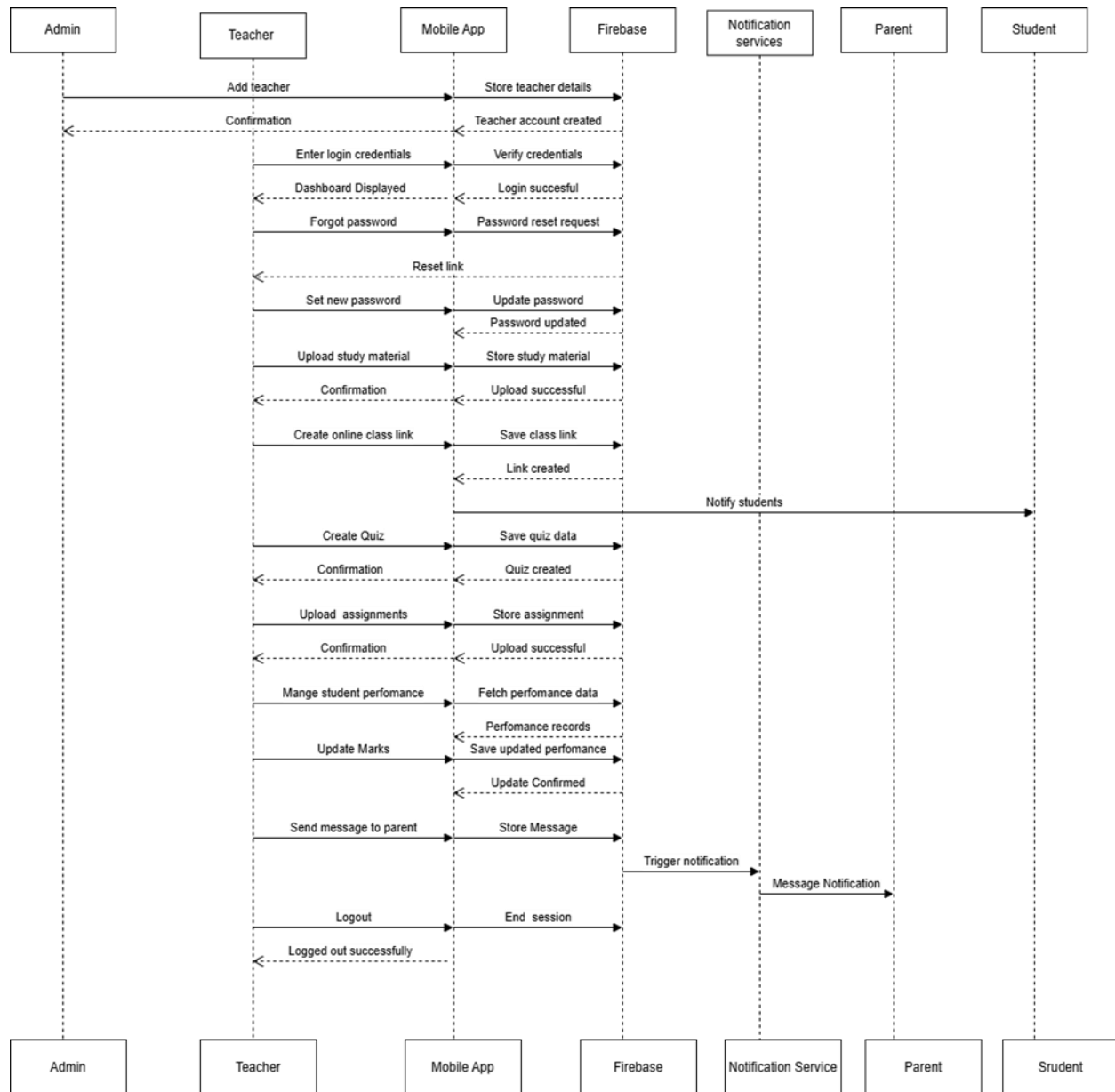


Figure 19: Sequence Diagram (Teacher)



## Parent Sequence diagram



Figure 20: Sequence Diagram (Parents)



## 2.State Transition Diagram

Shows state transitions within the software, especially useful for systems with complex workflows or lifecycle states.

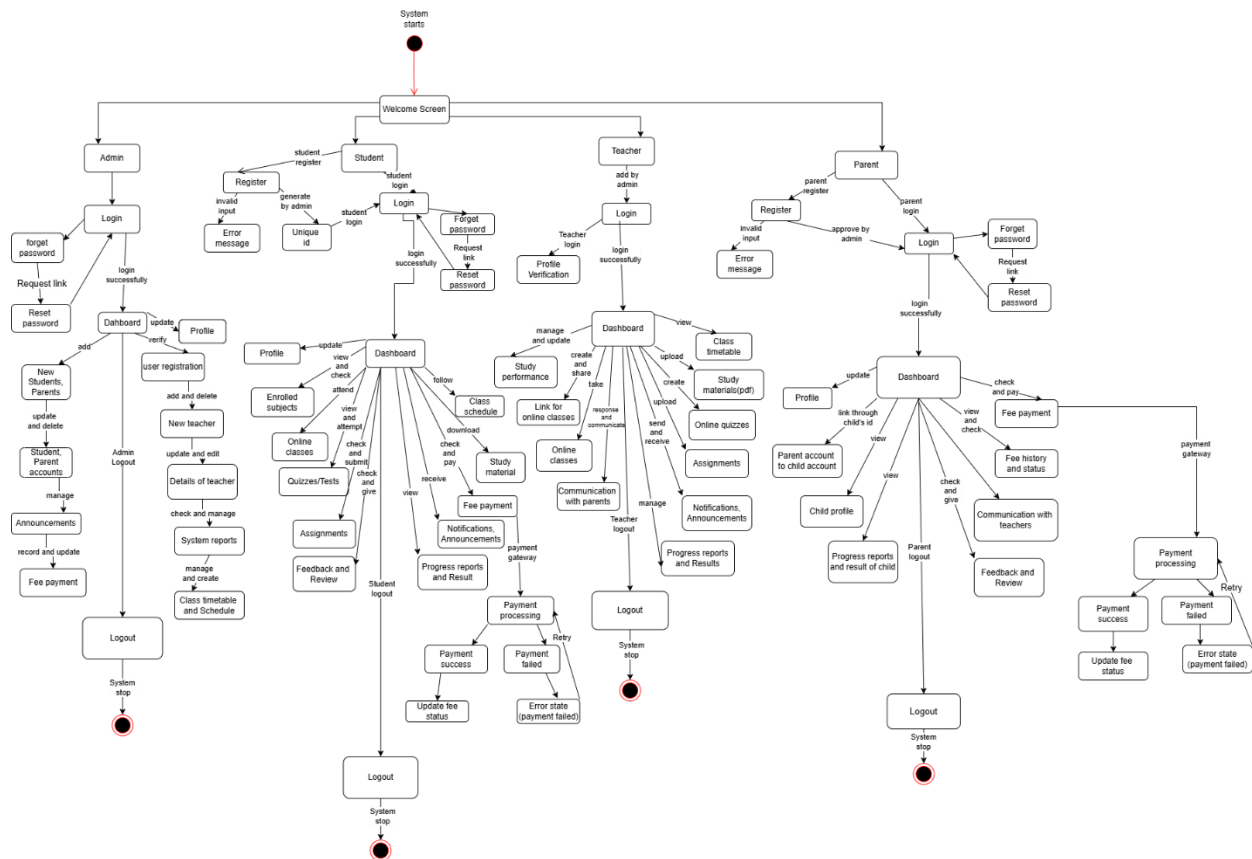


Figure 21: State Diagram

## 4.8 Design Decisions

We made key design decisions to ensure our system easy to use:

### 1. Object Oriented Design (OOD)

We use Object- Oriented Programming (OOP) to organize the system in modular and maintainable way. This makes the system easy to upgrade, scale and manage. The KIPS COACHING management system is build using an object – oriented approach, where each object represents real –world entities within the coaching environment, such as Students, Teachers, Subjects,





Assignments and notifications. Each object contains relevant attributes and methods, making the system organized and efficient for future enhancements.

Table 66: Object and attributes

| Object Name  | Description                                                                    | Attributes                                                                                   |
|--------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Admin        | Manage overall system including students, teachers, parents, subjects and fees | name, email, password                                                                        |
| Student      | Represent a students enrolled in KIPS Coaching Kharian                         | student_id, name, class, parent_contact                                                      |
| Teacher      | Represents a teacher or faculty member                                         | teacher_id, name, password, email                                                            |
| Parent       | Represent a parents                                                            | Parent_id, name, password, email                                                             |
| Subjects     | Represents subjects offered by the coaching center                             | subject_id, name, teacher_assigned_id                                                        |
| Live class   | Represents live classes conducted by the teachers                              | class_id, subject_id, teacher_id, date, time, meeting link, platform, duration               |
| Quiz         | Represents quizzes created by the teacher                                      | quiz_id, subject_id, teacher_id, duration, total marks, topics                               |
| Assignment   | Represents assignments for students                                            | assignment_id, subject_id, teacher_id, student_id, due_date, submission_date, marks_obtained |
| Notification | System generated notifications for the students, teacher and parent            | notification_id, recipient_type, recipient_id, message                                       |
| Fee          | Track student fee payment                                                      | fee_id, student_id, amount, due_date, payment_date, payment_method, status                   |
| Timetable    | Represents daily or weekly class schedules                                     | timetable_id, class_id, day, start_time, end_time                                            |



## 2.Model-View-Controller (MVC) Architecture

We followed the MVC pattern to separate data, business logic and the user interface. This approach keeps the system organized, easy to maintain and user friendly for students, teachers and admins.

## 3.Database Design and Normalization

We use firebase as the backend database to ensure fast, reliable data storage. The data is normalized to avoid duplication, ensure quick searches and maintain consistency for student records, study materials and fee transactions.

## Conclusion

By using OOP, MVC and an optimized database, our KIPS Coaching Kharian provides a smooth and user friendly interface.

Students can easily view subjects, attend online classes, view assignments and fee payments. Teacher can efficiently manage study materials, assignments, quizzes and student performance. Over all the system ensures efficiency, reliability and ease of use for everyone involved.

## 4.9Summary

This chapter explain how the **KIPS COACHING KHARIAN App** provides a smooth and automated academic management system. The system follows an object- oriented approach, making it well- structures and easy to manage. Using an object oriented approach, the app efficiently manages student records, class schedules, assignments, notifications and fee details. The sequence and state diagram showed how the student, teacher, parent and admin modules interact and how task moves through different stages. Key improvements such as fee alerts, parent monitoring and teacher tools for the uploading quizzes and assignments make the app easy to use, efficient and reliable.

## 4.10 References

- KIPS Education System

KIPS Virtual Learning Platform.

<https://kips.edu.pk>

Related system reference (Student enrolments and recorded lectures).

- The Educators School System

Online Academic Management System.

<https://educators.edu.pk>

Example of integrated digital education platform.

- Roots International Schools

Roots Learning Management System.

<https://rootsinternational.edu.pk>

Reference for paid and large-scale LMS platforms.



- KIPS Preparations Kharian Campus

KIPS Learning Management System.

<https://kipsprep.com/>

Reference for coaching and preparatory education platform.